1st Order DEs: Integrating Factors

Questions 1–45

Question 1. Find

a)
$$\frac{\mathrm{d}}{\mathrm{d}x}(6xy)$$

b)
$$\frac{\mathrm{d}}{\mathrm{d}x}(e^{3x}y)$$

c)
$$\frac{\mathrm{d}}{\mathrm{d}x}(y\tan 4x)$$

d)
$$\frac{\mathrm{d}}{\mathrm{d}x}(y\ln 2x)$$

Question 2.

a) Find
$$\frac{\mathrm{d}}{\mathrm{d}x}(yx^2)$$

b) Find the general solution of each of the following differential equations

(i)
$$x^2\frac{\mathrm{d}y}{\mathrm{d}x}+2xy=4$$

(ii)
$$x^2\frac{\mathrm{d}y}{\mathrm{d}x}+2xy=4x$$

Question 3.

a) Find

$$\frac{\mathrm{d}}{\mathrm{d}x}(y \sin x)$$

b) Find the general solution of each of the following differential equations

(i)

$$\sin x \frac{\mathrm{d}y}{\mathrm{d}x} + y \cos x = 3x^2$$

(ii)

$$\sin x \frac{\mathrm{d}y}{\mathrm{d}x} + y \cos x = e^x$$

Question 4.

a) Find

$$\frac{\mathrm{d}}{\mathrm{d}x}(e^{5x} y)$$

b) Find the general solution of the differential equation

$$e^{5x} \frac{\mathrm{d}y}{\mathrm{d}x} + 5e^{5x} y = 6e^{3x}$$

c) Hence, given that $y = 0$ when $x = \ln 2$, show that the particular solution of this differential equation can be written as

$$y = \frac{2e^{3x} - 16}{e^{5x}}$$

Question 5. The following differential equations are such that the left-hand side can be written in the form

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mu(x)y)$$

without the need to rearrange and find an integrating factor. Find the general solution to each of the differential equations

a)

$$x^4\frac{\mathrm{d}y}{\mathrm{d}x} + 4x^3y = x^8$$

b)

$$\cos 2x\frac{\mathrm{d}y}{\mathrm{d}x} - 2y\sin 2x = \cos 2x$$

c)

$$\tan 3x\frac{\mathrm{d}y}{\mathrm{d}x} + 3y\sec^2 3x = \cos 3x$$

d)

$$e^{-9x}\frac{\mathrm{d}y}{\mathrm{d}x} - 9e^{-9x}y = e^{-8x}$$

Question 6. Find and simplify

a)

$$e^{\int 2x\mathrm{d}x}$$

b)

$$e^{\int \frac{1}{x}\mathrm{d}x}$$

c)

$$e^{\int \frac{1}{2x+1}\mathrm{d}x}$$

d)

$$e^{\int \tan x\mathrm{d}x}$$

Question 7. For each of the following differential equations, find the integrating factor. You are not required to solve the differential equation.

a)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + 8y = e^x$$

b)
$$\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = x$$

c)
$$\frac{\mathrm{d}y}{\mathrm{d}x} - xy = 4$$

d)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + (2x+1)y = x^2$$

Question 8. It is given that
$$x^3\frac{\mathrm{d}y}{\mathrm{d}x} + x^3y = \frac{2}{e^x}$$

a) Show that an integrating factor is e^x
b) Solve the differential equation using the integrating factor in a).
c) Solve the differential equation using the following integrating factors. In each case, explain why your answer is equivalent to the answer given in b).
   (i) $2e^x$
   (ii) e^{x+2}

Question 9. It is given that

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$$

where $\mathrm{P}(x)$ and $\mathrm{Q}(x)$ are continuous functions of x. Suppose that the equation is multiplied by a function $\mu(x)$ so that the left-hand side becomes the derivative of a product:

$$\mu(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mu(x)\mathrm{P}(x)y = \frac{\mathrm{d}}{\mathrm{d}x}(\mu(x)y)$$

a) Use the product rule to show that $\mu(x)$ must satisfy

$$\mu'(x) = \mathrm{P}(x)\mu(x)$$

b) Hence show that a suitable integrating factor is

$$\mu(x) = e^{\int \mathrm{P}(x)\mathrm{d}x}$$

Question 10. Archie is attempting to solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = e^{3x}$$

His attempt is shown below: Integrating Factor $= e^{\int -3\mathrm{d}x} = -3e^{-3x}$
Multiplying the equation by $-3e^{-3x}$:

$$-3e^{-3x}\frac{\mathrm{d}y}{\mathrm{d}x} + 9e^{-3x}y = -3$$

This gives: $\frac{\mathrm{d}}{\mathrm{d}x}(-3e^{-3x}y) = -3$
Integrating: $-3e^{-3x}y = -3x + c$
Rearranging: $y = e^{3x}\left(x - \frac{c}{3}\right)$

a) Identify the error in Archie's work.
b) Verify whether Archie's integrating factor $-3e^{-3x}$ is valid
c) A different student uses the integrating factor e^{-3x} and obtains the solution $y = e^{3x}(x + c)$. Explain why this solution is equivalent to the solution found in Archie's work.

Question 11. Find the general solution for each of the following differential equations:

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 10y = e^{2x}$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 7y = 8$$

c)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 8y = 100$$

d)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 20y = 6e^{-12x}$$

Question 12. Simplify the following:

a)

$$e^{\ln x}$$

b)

$$e^{\ln x^2}$$

c)

$$e^{4\ln x}$$

d)

$$e^{-3\ln x}$$

Question 13. Find the general solution for each of the following differential equations:

a)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} = x$$

b)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} = 30$$

c)
$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{3y}{x} = x^3$$

d)
$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{5y}{x} = \frac{40}{x^4}$$

Question 14. Find the general solution for each of the following differential equations:

a)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x+1} = 6x$$

b)
$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x-3} = 12$$

c)
$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4y}{2x+1} = 2$$

Question 15. It is given that

$$p\frac{\mathrm{d}y}{\mathrm{d}x} + qy = r$$

where p, q and r are positive constants. It is also given that $y = \frac{2r}{q}$ when $x = 0$

a) Find y in terms of x

b) Find the value of y as $x \to \infty$

Question 16. Find the general solution to

a)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 1$$

b)

$$(x+1)\frac{\mathrm{d}y}{\mathrm{d}x} + y = 2$$

c)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = 1$$

d)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = 2x^3$$

Question 17. Find the general solution to

a)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + 3y = \sqrt{x}$$

b)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = \frac{1}{\sqrt{x}}$$

Question 18. Find the general solution to

a)

$$2x^{\frac{5}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} + x^{\frac{3}{2}}y = 2$$

b)

$$2x^{\frac{5}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} - x^{\frac{3}{2}}y = 2$$

Question 19. It is given that $x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 7xy = \frac{2}{x^5}$ where $x > 0$. Given that $y = 1$ when $x = 2$, show that y is a decreasing function.

Question 20.

a) Find the general solution of the differential equation $\sin x\frac{\mathrm{d}y}{\mathrm{d}x} - y\cos x = 2^x \sin^2 x$ where $x \in \mathbb{R}$

b) Hence find the particular solution of this differential equation, given that $y = 0$ when $x = \frac{\pi}{2}$

Question 21. The following questions in this section require the use of A-Level Maths Calculus techniques, including Reversing the Chain Rule, Double Angle Formulae, Integration by Parts
Find the general solution for each of the following differential equations:

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + xy = x$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 6x^3 y = 8x^3$$

Question 22. Find the general solution for each of the following differential equations:

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y\cos x = \cos x$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - y\sin 2x = 4\sin 2x$$

Question 23. Find the general solution to

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} = 6\sqrt{x^3 + 2}$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} = \frac{6}{\sqrt{x^4 + 2}}$$

Question 24. It is given that the curve that satisfies the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{6y}{3x - 2} = 54(3x - 2)^6$$

has a stationary point when $x = \frac{1}{3}$
Find the equation of the curve.

Question 25. Find y in terms of x, given that

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = 36x^6 \ln x$$

and $y = 2$ when $x = 1$

b)

$$x\sqrt{x}\frac{\mathrm{d}y}{\mathrm{d}x} + y\sqrt{x} = 8x\sqrt{x^3 + 4x}$$

and $y = 1$ when $x = \sqrt{5}$

c)

$$\tan x \frac{\mathrm{d}y}{\mathrm{d}x} + y \sec^2 x = \tan^2 x$$

and $y = 1$ when $x = \frac{\pi}{4}$

Question 26. The gradient at any point (x, y) on a curve C is proportional to $(y - x)$. The curve passes through $(1, 3)$ and the gradient of the curve at this point is -8. Find the equation of the curve.

Question 27.

a) Find the general solution of the differential equation

$$(1-\cos 2x)\frac{\mathrm{d}y}{\mathrm{d}x}+2y\sin 2x=2(\sin x+\cos x)$$

b) Hence, given that $y=5$ when $x=\frac{\pi}{2}$, show that the particular solution of this differential equation can be written as

$$y=\frac{\sin x-\cos x+4}{\sin^2 x}$$

Question 28.

a) Find the general solution of the differential equation

$$x\frac{\mathrm{d}y}{\mathrm{d}x}+(2x+1)y=xe^x$$

b) Hence, given that $y=\frac{1}{3}$ when $x=1$, show that the particular solution of this differential equation can be written as

$$y=\frac{(3x-1)e^x+e^{3-2x}}{9x}$$

Question 29. Given that

$$\frac{\mathrm{d}y}{\mathrm{d}x} - y\tan x = \sin 2x$$

where $-\frac{\pi}{2} < x < \frac{\pi}{2}$, and $y = \frac{4}{3}$ when $x = 0$, find y in terms of x

Question 30.

a) Find the general solution of the differential equation

$$\sec x\frac{\mathrm{d}y}{\mathrm{d}x} + y\operatorname{cosec} x = 2\tan x$$

where $0 < x < \frac{\pi}{2}$

b) Hence find the particular solution of this differential equation, given that $y = \frac{12}{\sqrt{3}}$ when $x = \frac{\pi}{3}$

Question 31. Given that

$$x^4\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{x^3y}{\ln x}=3$$

where $x>1$, and $y=e^{-3}$ when $x=e$, show that

$$y=\frac{7e^{-3}x^3-3\ln x-1}{3x^3\ln x}$$

Question 32. The following questions in this section require the use of A-Level Further Maths Calculus techniques, including Partial Fractions, Inverse Trigonometric Functions, Hyperbolic Functions, Inverse Hyperbolic Functions

Find the general solution for each of the following differential equations, giving your answer in the form $y=\mathrm{f}(x)$:

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x}+y\sinh x=2\sinh x$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x}-2y\cosh 3x=6\cosh 3x$$

c)

$$2\cosh^2 x\frac{\mathrm{d}y}{\mathrm{d}x}+y\sinh 2x=\sinh^5 x\cosh^2 x$$

Question 33. Find the general solution of the differential equation

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = \frac{7-x}{(x^2+7)(x+1)}$$

where $x \neq 0$ and $x \neq -1$

Question 34.

a) Show that

$$e^{\int \frac{1}{1-x^2}\,\mathrm{d}x} = \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$$

It is given that

$$(1-x^2)\frac{\mathrm{d}y}{\mathrm{d}x} + y = (1-x)^{\frac{3}{2}}(1+x)$$

where $-1 < x < 1$

b) Find the general solution of the differential equation, given that $y = \frac{2}{3}$ when $x = 0$

Question 35. Given that

$$(9-4x^2)\frac{\mathrm{d}y}{\mathrm{d}x}-4xy=2$$

where $-\frac{3}{2}<x<\frac{3}{2}$, and $y=0$ when $x=\frac{3\sqrt{2}}{4}$, find y in terms of x

Question 36. Given that

$$2x\frac{\mathrm{d}y}{\mathrm{d}x}-y=\sqrt{\frac{4x^3}{4x-x^2-1}}$$

where $2-\sqrt{3}<x<2+\sqrt{3}$, and $y=\sqrt{\frac{7}{2}}$ when $x=\frac{7}{2}$, find y in terms of x

Question 37. Prerequisite knowledge for Q37-Q38: Complementary Functions for solving Non-Homogeneous Differential Equations

Solve the following differential equations by finding the complementary solution and a particular integral.

a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = e^{5x}$$

b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = x$$

c)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 17\sin x$$

d)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 7y = e^{7x}$$

Question 38.

a) Show that $y = Ax + B + C\sin x + D\cos x$ is a particular integral of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 15\cos x - 4x$$

b) Hence find the general solution of the differential equation.

Substitutions

Question 39. It is given that

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y = xy^2$$

where $x > 0$. Use the substitution $u = \frac{1}{y}$ to find the general solution for y in terms of x

Question 40. It is given that

$$5xy^4\frac{\mathrm{d}y}{\mathrm{d}x} + 3y^5 = \frac{\sin 2x}{x^2}$$

where $x > 0$. Use the substitution $u = y^5$ to find the general solution for y in terms of x

Question 41. It is given that

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 4xy^3 = y$$

Given that $y = \frac{1}{3}$ when $x = 0$, use the substitution $u = \frac{1}{y^2}$ to find a particular solution for y^2 in terms of x

Question 42. It is given that

$$x^3 e^y \frac{\mathrm{d}y}{\mathrm{d}x} = 6x^2 e^x - 3x^2 e^y$$

Given that $y = \ln 6$ when $x = 1$, use the substitution $u = e^y$ to find a particular solution for y in terms of x

Question 43. It is given that

$$x\frac{\mathrm{d}y}{\mathrm{d}x} = y(\ln y + x)$$

where $x > 0$ and $y > 0$. Use the substitution $u = \ln y$ to find the general solution for y in terms of x

Question 44. It is given that

$$\frac{\mathrm{d}y}{\mathrm{d}x} = e^x \cos^2 y + \sin y \cos y$$

where $x \in \mathbb{R}$. Use the substitution $u = \tan y$ to find the general solution for y in terms of x

Question 45. For $x > 0$ and $y > 0$, it is given that

$$p\frac{\mathrm{d}y}{\mathrm{d}x} + qy = ry^{1-n}$$

where p, q, r and n are positive constants.

a) Use the substitution $u = y^n$ to find the general solution for y in terms of x

b) Find the value of y as $x \to \infty$ in terms of the constants q, r and n

1st Order DEs: Sketching Solutions

Questions 1–16

Question 1. Sketch the graph of $y = x^3 + c$ for $c = -1, c = 0$, and $c = 1$ on a single set of axes.

Question 2. Sketch the graph of $y = \frac{1}{x} + c$ for $c = -1, c = 0$, and $c = 1$ on a single set of axes.

$\frac{dy}{dx} = \mathrm{f}(x)$

Question 3. It is given that

$$\frac{dy}{dx} = 2x$$

a) For what value of x is the gradient zero?
b) Find the general solution to the differential equation.
c) Sketch the family of solutions represented by the general solution on a single set of axes.
d) Describe how the family of solutions are related geometrically.

Question 4. It is given that

$$\frac{dy}{dx} = \cos x$$

a) Find the general solution to the differential equation.
b) Sketch the family of solutions represented by the general solution on a single set of axes.

Separation of Variables

Question 5. It is given that

$$\frac{dy}{dx} = 3y$$

a) Show that $y = Ae^{3x}$, where A is a constant.
b) Sketch the family of solutions represented by the general solution when $A < 0$ and when $A > 0$ on a single set of axes.
c) Describe what happens as $x \to -\infty$

Question 6. It is given that

$$\frac{dy}{dx} = y - 2$$

a) Determine the constant solution to the differential equation.
b) Find the general solution to the differential equation.
c) Sketch the family of solutions represented by the general solution on a single set of axes.

Question 7. Solve each of these differential equations and hence sketch the family of solutions.

a) $\frac{dy}{dx} = -2y$

b) $\frac{dy}{dx} = 1 - y$

c) $\frac{dy}{dx} = y^2$

d) $\frac{dy}{dx} = \frac{1}{2y}, \quad y > 0$

Prerequisite Knowledge for Q8: A-Level Maths Partial Fractions

Question 8. It is given that

$$\frac{dy}{dx} = y(1 - y)$$

a) Determine the constant solutions to the differential equation.

b) Use separation of variables to show that

$$y = \frac{Ae^x}{1 + Ae^x}$$

c) Explain why one of the constant solutions from part a) does not appear in the solution found in part b).

d) Show that the solution in b) can also be written in the form

$$y = \frac{1}{1 + Be^{-x}}$$

e) Explain why the other constant solution from part a) does not appear in the solution found in part d).

f) Sketch the family of solutions represented by the solutions in parts a) and b) on a single set of axes.

Integrating Factors

Question 9. It is given that

$$\frac{dy}{dx} + y = 3$$

a) Find the constant solution of the differential equation.
b) Use an integrating factor to solve the differential equation.
c) Sketch the family of solutions on a single set of axes.

Question 10. It is given that

$$\frac{dy}{dx} - y = 2$$

a) Find the constant solution of the differential equation.
b) Solve the differential equation.
c) Describe the behaviour of solutions as $x \to \infty$
d) Sketch the family of solutions on a single set of axes.

Question 11. It is given that

$$\frac{dy}{dx} + 2y = 6$$

a) Find the constant solution of the differential equation.
b) Solve the differential equation.
c) Describe the behaviour of solutions as $x \to \infty$
d) Sketch the family of solutions on a single set of axes.

Prerequisite Knowledge for Q12-Q15: Reversing the Chain Rule, Integration by Parts

Question 12. It is given that

$$\frac{dy}{dx} + 2xy = 4x$$

a) Solve the differential equation.
b) Describe the behaviour of the solutions as $x \to \infty$ and as $x \to -\infty$
c) Sketch the family of solutions on a single set of axes.

Question 13. It is given that

$$\frac{dy}{dx} + y = x + 1$$

a) Use an integrating factor to show that the general solution is

$$y = x + Ae^{-x}$$

where A is a constant.

b) Show that the line $y = x$ is itself a solution of the differential equation.

c) Explain why every solution curve approaches the line $y = x$ as $x \to \infty$

d) Sketch the family of solutions on a single set of axes.

e) Show that distinct solution curves do not intersect.

Question 14. It is given that

$$\frac{dy}{dx} - y = x$$

a) Solve the differential equation.

b) Show that the line $y = -x - 1$ is a solution of the differential equation.

c) Describe the behaviour of the solutions as $x \to -\infty$

d) Describe the behaviour of the solutions as $x \to \infty$

e) Sketch the family of solutions on a single set of axes.

Question 15. It is given that

$$\frac{dy}{dx} + y = \sin x + \cos x$$

a) Solve the differential equation to show that $y = \sin x + Ae^{-x}$ for constant A
b) Explain why every solution approaches the curve $y = \sin x$ as $x \to \infty$
c) Sketch the family of solutions on a single set of axes.

Prerequisite Knowledge for Q16: Calculus with Hyperbolic Functions

Question 16. It is given that

$$\frac{dy}{dx} + y = 4\sinh x$$

a) Solve the differential equation to show that

$$y = e^x + (A - 2x)e^{-x}$$

where A is a constant.
b) State the coordinates of the point where a solution curve crosses the y-axis in terms of A
c) Describe the behaviour of the solutions as $x \to \infty$ and as $x \to -\infty$
d) Show that the stationary points of the family of solution lie on the curve

$$y = 4\sinh x$$

e) Sketch the family of solutions on a single set of axes.

1st Order DEs: Modelling

Questions 1–46

Question 1. A bucket contains V litres of water at time t minutes. Water flows into the bucket at a constant rate. Write down a differential equation for:

a) $\frac{dV}{dt}$

b) A bucket contains V litres of water at time t minutes. The bucket leaks at a rate proportional to the volume of water present. Write down a differential equation for $\frac{dV}{dt}$.

c) A tank contains x litres of water at time t minutes. Water is pumped into the tank at a constant rate of 6 litres per minute. Water leaks out at a rate proportional to the amount of water in the tank. Write down a differential equation for $\frac{dx}{dt}$.

d) A cistern is fitted with a control valve. Let V be the volume of water (in litres) at time t minutes. The valve acts so that the rate of change of V is proportional to the difference between 200 litres and V. Write down a differential equation for $\frac{dV}{dt}$.

e) A metal object has temperature T°C at time t minutes. It is placed in a room at a constant temperature of 18°C. The object cools at a rate proportional to the difference between its temperature and the room temperature. Write down a differential equation for $\frac{dT}{dt}$.

f) A lake contains P tonnes of pollutant at time t months. Pollutant enters the lake at a rate of $0.5t$ tonnes per month. Natural processes remove pollutant at a rate proportional to the amount present. Write down a differential equation for $\frac{dP}{dt}$.

Question 2. A cart of mass 6 kg moves along a straight horizontal track. At time t seconds its speed is v m s^{-1}. A motor provides a forward driving force of magnitude $(18 - 3t)$ N. A resistive force of magnitude $4v$ N acts opposite to the motion. Assuming the cart continues moving forwards, form a differential equation for v in terms of t.

Question 3. A sled of mass 10 kg is pulled along level snow. At time t seconds its velocity is v m s^{-1}. The pulling force is constant and equal to 25 N. The resistive force from the snow is modelled as having magnitude $(5 + 0.2t)v$ N, acting opposite to the direction of motion. Form a differential equation for v in terms of t.

Question 4. A skydiver of mass 80 kg jumps from a stationary helicopter and falls vertically downwards. At time t seconds, their speed is v m s^{-1}. The air resistance is modelled as a force of magnitude $6v$ N, opposing the direction of motion. Assume acceleration due to gravity is g m s^{-2}. Form a differential equation relating v and t.

Question 5. A particle of mass 2 kg moves along a straight horizontal line. At time t seconds, its velocity is v m s^{-1}. The particle is acted on by a driving force of magnitude $6e^{-t}$ N in the direction of motion, and a resistive force of magnitude v^2 N acting opposite to the motion.

a) Form a differential equation for v in terms of t.

b) Explain why the differential equation cannot be solved using the integrating factor method.

Question 6. A particle of mass 5 kg moves along a straight horizontal line. At time t seconds, its velocity is v m s^{-1}. The particle is acted on by a forward driving force of magnitude 8 N, and a resistive force of magnitude $2v\,\frac{dv}{dt}$ N acting opposite to the direction of motion.

a) Form a differential equation for v in terms of t.

b) Explain why the differential equation cannot be solved using the integrating factor method.

It is given that the initial velocity of the particle is 0.5 m s^{-1}.

c) Solve the differential equation, writing v in terms of t.

Question 7. A tank contains 80 litres of water which has 12 kg of salt dissolved into it. Find the concentration of salt in kg l^{-1}:

a) in the tank;

b) in terms of S and V;

c) A tank contains 200 litres of water which has 5 kg of salt dissolved into it. Liquid flows out of the tank at 3 litres per minute. Write down the rate (in kg/min) at which the salt leaves.

Question 8. A tank contains 100 litres of solution with S kg of salt dissolved into it. The mixture is stirred. Liquid flows out at 4 litres per minute. Write down the rate at which salt leaves the tank in terms of S.

Question 9. Brine enters a tank at 5 litres per minute with concentration 0.3 kg per litre. The mixture leaves at 5 litres per minute. Let S kg be the mass of salt in the tank. The volume of the tank is 100 litres.

a) Find the rate at which salt enters the tank.

b) Write the rate at which salt leaves the tank in terms of S.

c) Hence form a differential equation for $\frac{dS}{dt}$.

Question 10. A tank contains 50 litres of solution and initially 2 kg of salt is in the solution. Brine enters at 2 litres per minute with concentration 0.4 kg per litre, and the mixture leaves at 2 litres per minute. Let S kg be the mass of salt after t minutes. Form a differential equation for $\frac{dS}{dt}$ and state an initial condition for S.

Question 11. A tank initially contains 100 litres of water. Water flows into the tank at a rate of 3 litres per minute and flows out of the tank at a rate of 5 litres per minute. Write down an equation for the volume, V litres, of the water at time t minutes.

Question 12. A tank initially contains 60 litres of water. Water flows into the tank at a rate of 7 litres per minute and flows out of the tank at a rate of 2 litres per minute. Write down an equation for the volume, V litres, of the water at time t minutes, and state a suitable domain.

Question 13. A tank initially contains V_0 litres of water. Water flows into the tank at a rate of p litres per minute and out of the tank at a rate of q litres per minute. Write down an equation for the volume, V litres, of the water at time t minutes.

Question 14. A tank contains 100 litres of solution with S kg of salt dissolved into it. Pure water enters the tank at a rate of 4 litres per minute and the mixture leaves the tank at a rate of 4 litres per minute. Write down a differential equation for $\frac{dS}{dt}$.

Question 15. A tank initially contains 100 litres of solution with S kg of salt mixed into it. Pure water enters the tank at a rate of 4 litres per minute and the mixture leaves the tank at a rate of 6 litres per minute.

a) Find an equation for V, the volume of salt solution, in the tank at time t minutes.

b) Write down the rate at which salt leaves the tank.

c) Hence form a differential equation for $\frac{dS}{dt}$.

Question 16. A tank initially contains 100 litres of solution with 6 kg of salt dissolved in it. Brine of concentration 0.2 kg per litre enters the tank at 4 litres per minute, and the mixture leaves the tank at 6 litres per minute. Let S kg be the mass of salt in the tank after t minutes. Form a differential equation for $\frac{dS}{dt}$.

Question 17. A tank initially contains 40 litres of solution with 5 kg of salt dissolved into it. Brine of concentration 0.5 kg per litre enters the tank at 3 litres per minute and flows out at 2 litres per minute. Let S kg be the mass of salt in the tank after t minutes.

a) Find an equation for V, the volume of solution in the tank at time t minutes.

b) Form a differential equation for $\frac{dS}{dt}$.

c) Explain briefly what happens to the concentration as $t \to \infty$.

Question 18. A student models the temperature T°C of a cup of tea after t minutes using $\frac{dT}{dt} = -0.12(T - 18)$.

a) Explain what the constant 18 represents.

b) Explain what the term $(T - 18)$ represents.

c) Explain why the negative sign is sensible.

d) Explain what the constant 0.12 represents.

Question 19. A population of fish, P, is measured in thousands. Time t is measured in years. A student models the population using $\frac{dP}{dt} = 0.4P - 2$.

a) Explain what the term $0.4P$ represents.

b) Explain what the term -2 represents.

c) Explain why the term -2 does not depend on P.

Question 20. The amount of charge stored in a capacitor at time t seconds is Q, measured in coulombs (C). The rate at which charge is added is called the charging rate and is given by $\frac{dQ}{dt} = 6 - \frac{Q}{4}$. A student models the charging process using:

a) Explain the term 6 represents.

b) Explain what the term $\frac{Q}{4}$ represents.

c) Explain why the sign of the $\frac{Q}{4}$ term makes sense physically.

d) Explain what happens to $\frac{dQ}{dt}$ as Q becomes large.

Question 21. A tank contains 100 litres of liquid, which is always well mixed. Let S kg be the mass of salt in the tank after t minutes. It is given that S satisfies $\frac{dS}{dt} = 0.8 - \frac{5S}{100}$.

a) What does the term 0.8 represent?

b) What does the term $\frac{5S}{100}$ represent?

c) State the outflow rate in litres per minute.

d) State the inflow concentration of salt in kg per litre.

Question 22. A gas tank contains a mixture of gases. Let x be the volume of oxygen (cm^3) after t hours. Suppose x satisfies $\frac{dx}{dt} = 3.5 - \frac{2x}{50+4t}$.

a) What does 3.5 represent?

b) Explain why the term $\frac{2x}{50+4t}$ represents oxygen leaving the tank.

c) Find the time at which the volume of oxygen is decreasing at a rate of 0.5 cm^3 h^{-1}. Give your answer in minutes.

Question 23. A well-mixed tank contains salt solution of constant volume 40 litres. Solution flows out of the tank at a rate of 3 litres per minute. Let S be the mass of salt in the tank, measured in kg, after t minutes. A student proposes the model $\frac{dS}{dt} = 0.6 - \frac{3S}{40}$.

a) State the units of $\frac{dS}{dt}$.

b) State the units of the constant term 0.6.

c) Show that the term $\frac{3S}{40}$ has the correct units.

Question 24. A tank contains a liquid of volume V litres at time t minutes. A chemical is dissolved in the liquid. Let x be the mass of the chemical in the tank, measured in grams. The volume of liquid in the tank is modelled by $V = 80 + 2t$. A student writes the differential equation $\frac{dx}{dt} = 1.5 - \frac{4x}{80+2t}$.

a) The student claims that the constant term 1.5 represents the chemical entering the tank. What must the units of 1.5 be?

b) Determine the units of the number 4.

c) Explain why the equation is dimensionally inconsistent if x were measured in kg instead of grams, unless the constant term were adjusted.

Question 25. A population of bacteria is being studied. Let P be the population at time t hours, measured in thousands of bacteria. A student proposes the model $\frac{dP}{dt} = k - \frac{P}{12+t}$.

a) State the units of $\frac{dP}{dt}$.

b) Hence state the units of k.

c) Explain why the number 12 must have units, and state its units.

d) The student later changes the time unit so that t is measured in minutes instead of hours. Write the model in terms of the new variable T, where T is time in minutes. (You may leave k in terms of its original units but the equation must be dimensionally consistent.)

Question 26. A tank initially contains 90 litres of water. Water drains from the tank at a constant rate of 3 litres per minute. Let V be the volume of water (in litres) after t minutes.

a) Write down an expression for V in terms of t.

b) State the domain of t for which the model is physically valid.

c) Explain why the model is not valid outside this domain.

Question 27. A tank initially contains 40 litres of water. Water flows into the tank at 6 litres per minute and flows out of the tank at 2 litres per minute. The tank has a maximum capacity of 120 litres. Let V be the volume (in litres) of liquid in the tank after t minutes.

a) Find $V(t)$.

b) Find the time when the tank begins to overflow.

c) Hence state the domain of t for which the model is valid.

d) Give one limitation of the model.

Question 28. A population of rabbits is modelled by $\frac{dP}{dt} = 0.12P$, where P is the population size after t years.

a) Solve the differential equation, given that $P(0) = 500$.

b) Use your model to predict the population after 20 years, giving your answer to three significant figures.

c) Explain why this prediction may not be realistic.

d) State one improvement that could be made to the model.

Question 29. A cup of coffee is modelled by $\frac{dT}{dt} = -0.18(T - 20)$, where T is the temperature of the coffee in °C and t is the time in minutes.

a) Explain why the model predicts that T will never be below 20°C.

b) Solve the differential equation, given that $T = 70$ when $t = 0$.

c) State two assumptions made in this model.

Question 30. A chemical is placed in a container and decays according to the model $\frac{dM}{dt} = -\frac{5M}{100-t}$, where M is the mass of chemical (in grams) at time t minutes.

a) State the domain of t for which the model is valid.

b) Explain why the model cannot be valid for all $t \geq 0$.

c) Solve the differential equation, given that $M(0) = 80$.

d) In the domain of t for which the model is valid, use your solution to describe what happens to M as $t \to 100$.

e) Suggest one reason why the model might be unrealistic.

Question 31. A tank contains 120 litres of water in which 10 kg of salt is dissolved. At time t minutes, brine containing 0.25 kg of salt per litre flows into the tank at a rate of 4 litres per minute. The mixture is kept uniform by stirring and flows out of the tank at a rate of 4 litres per minute. Let S kg be the mass of salt in the tank after t minutes.

a) Show that S satisfies the equation $\frac{dS}{dt} = 1 - \frac{S}{30}$.

b) Solve the differential equation.

c) Find the mass of salt in the tank after 20 minutes.

Question 32. A tank contains 50 litres of solution with 5.5 kg of salt dissolved in it. At time t minutes, brine containing 0.25 kg of salt per litre flows into the tank at a rate of 4 litres per minute. The mixture is stirred and flows out of the tank at a rate of 6 litres per minute. Let S kg be the mass of salt in the tank after t minutes.

a) Show that S satisfies the differential equation $\frac{dS}{dt} = 2 - \frac{3S}{25+t}$.

b) Solve the differential equation.

c) Find the concentration as $t \to \infty$.

d) State one limitation of the model.

Question 33. A pressure vessel initially contains 600 cm^3 of nitrogen. Gas leaks out of the vessel at a constant rate of 30 cm^3 per hour, and at the same time a gas mixture is pumped into the vessel at a rate of 70 cm^3 per hour. The gas being pumped in contains 8% carbon dioxide and 92% nitrogen. Assume the gases mix and uniformly inside the vessel. Let x cm^3 be the volume of carbon dioxide in the vessel.

a) Show that the situation can be modelled by the differential equation $\frac{dx}{dt} = 5.6 - \frac{3x}{600+40t}$.

b) Hence find the volume of carbon dioxide in the vessel after 5 hours, giving your answer to three significant figures.

c) Find the concentration (as a percentage) of carbon dioxide in the vessel as $t \to \infty$.

d) Describe one limitation of this model.

Question 34. Two different liquids, ethanol and water, are mixed together in a vessel. The contents are stirred continuously so that the mixture is always uniform. Initially, the vessel contains 15 litres of ethanol and 25 litres of water. From the time $t = 0$, ethanol is added to the vessel at a rate of 3 litres per second and water is added to the vessel at a rate of 1 litre per second. The well-mixed contents are removed at a rate of 4 litres per second. Hence the total volume in the vessel remains constant. Let x be the amount of ethanol in the vessel at time t seconds.

a) Show that x can be modelled by $\frac{dx}{dt} = 3 - \frac{x}{10}$.

b) Solve the differential equation to determine how long it will take for the ratio of ethanol to water in the vessel to be 2:1, according to the model. Give your answer to the nearest second.

c) It is observed experimentally that the solution takes a ratio of 2:1 of ethanol to water after 18 seconds. Use this information to comment on the validity of the model, giving a reason for your answer.

Question 35. A laboratory tank initially contains 5 litres of pure water. At time t minutes, a chemical solution is pumped into the tank at a constant rate of 100 ml per minute. At the same time, the well-mixed liquid leaves the tank through an outlet pipe at a constant rate of 50 ml per minute. The concentration of chemical in the incoming solution is not constant. It is modelled by $2 + 3e^{-0.08t}$ mg per litre. Let m be the mass of chemical (in mg) in the tank at time t minutes after pumping begins.

a) Find an expression for the volume of liquid in the tank, in litres, at time t.

b) Show that m satisfies a differential equation of the form $\frac{dm}{dt} + \frac{m}{100+t} = \frac{1}{10}(2 + 3e^{-0.08t})$.

Question 36. A metal rod is initially at temperature 0°C. Its temperature T at time t seconds is modelled by $\frac{dT}{dt} + T = 4e^t \cos t$. Find the temperature of the rod at time $t = \pi$, giving your answer in exact form.

Question 37. A company uses a subscription-based online service. Let N be the number of active subscribers at time t years.

a) The rate at which subscribers leave the service is proportional to the number of active subscribers. In one year, the number of subscribers lost is equal to 12% of the number at the start of the year. Assuming no new subscribers join, giving your answer to three significant figures, find the constant k such that $\frac{dN}{dt} = -kN$.

The company introduces a marketing campaign. The rate at which subscribers leave remains the same as in part a), it is also given new subscribers join at a rate of 30 per year and the company also gains 60 subscribers per year through a long-term partnership.

b) Write down a first-order differential equation for N in terms of t.

c) Initially, the company has 500 subscribers. Solve your differential equation from part b) to find N in terms of t.

d) Describe two simplifications that have been made in forming the model, that could affect its accuracy.

Question 38. A water treatment system is operating, and the mass of a chemical impurity, M grams, in the system is modelled by the differential equation $\left(\frac{4+2t}{\sqrt{t}}\right)\frac{dM}{dt} + \frac{4M}{\sqrt{t}} = 105(2+t)$, where t is the time in hours after the system starts operating. Initially, there are 4 grams of the impurity present.

a) Find, according to the model, the mass of impurity present after 6 hours.

b) Find, according to the model, the rate of change of the mass of impurity after 3 hours.

c) State one assumption of the model that may not be valid in practice.

Question 39. A pharmaceutical company is testing a drug in solution. Let x be the amount (in mg) of drug remaining in the solution after t minutes. The drug breaks down over time, and the rate of decrease of the drug in solution is proportional to $x^2 + 16$. Initially, the amount of the drug in the solution is 120 mg. After 50 minutes, this has reduced to 60 mg. Find the amount of drug remaining in the solution after 200 minutes.

Question 40. A particle of mass 3 kg moves along a straight horizontal line. At time t seconds, its velocity is v m s^{-1}. The particle is acted on by two forces: a driving force of magnitude $(6+2t)$ N acting in the direction of motion, and a resistive force of magnitude $3v$ N acting opposite to the motion.

a) By forming a differential equation, show that the motion of the particle satisfies $\frac{dv}{dt} + v = 2 + \frac{2}{3}t$.

Question 41. A small trolley of mass 0.4 kg moves along a straight horizontal track. At time t seconds, its speed is u m s^{-1}. Two horizontal forces act on the trolley: a driving force P N acting in the direction of motion, and a resistive force Q N opposing the direction of motion. In an initial model, P is assumed to be constant with a constant of proportionality k, and Q is assumed to be proportional to speed u with a constant of proportionality c, where $k, c > 0$.

a) Show that the situation can be modelled by $\frac{du}{dt} = \frac{5}{2}(k - cu)$.

b) Solve this differential equation to obtain u in terms of t, k, and c, given that the trolley starts from rest.

You are now given that $k = 1.6$ and $c = 0.8$.

c) Find the acceleration of the trolley when $t = 2$ seconds.

In a refined model, the driving force has constant magnitude 1.6 N, while the resistive force remains as it was in the original model.

d) Write down the differential equation satisfied by u in Model B.

e) By considering the sign of $\frac{du}{dt}$, explain why Model B predicts that u tends to a limiting value as $t \to \infty$, and find this limiting value.

Question 42. A particle of mass m kg is projected vertically downwards with initial speed u m s^{-1}. At time t seconds its velocity is v m s^{-1}, where v is taken as positive downwards. The acceleration due to gravity is g m s^{-2}.

a) In Model A, the resistive force is assumed to have magnitude kmv N, acting upwards, where k is a positive constant. Show that v satisfies $\frac{dv}{dt} + kv = g$.

b) Solve the differential equation to obtain v in terms of t.

c) Show that the particle approaches a terminal velocity, and state its value.

In Model B, the resistive force is instead assumed to have magnitude $km\sqrt{v}$ N, acting upwards.

d) Write down the differential equation satisfied by v in Model B.

e) By considering the sign of $\frac{dv}{dt}$, explain why Model B predicts that v tends to a limiting value as $t \to \infty$, and find this limiting value.

Question 43. A train carriage moves along a straight horizontal track. The total mass of the carriage is 17 tonnes. At time t seconds its velocity is v m s^{-1}, where $0 \leq t \leq 10$. A resistive force acts at all times, modelled as a force of magnitude $2.12v$ N, acting opposite to the direction of motion. The carriage starts from rest at $t = 0$. For the first part of the motion, the carriage is pulled forward by an engine. Later, the engine stops pulling, and the carriage is slowed down by an additional braking force. At $t = 10$, the carriage comes to rest.

a) State the initial magnitude of the resistive force.

From $t = 6$ to $t = 10$, the engine is no longer pulling. Instead, the buffer exerts a constant force k N opposite to the motion.

b) Form a differential equation satisfied by v for $6 \leq t \leq 10$.

c) Solve your differential equation from part b) to find v in terms of t.

d) Hence find the speed of the carriage at $t = 6$.

e) Find the time (in minutes) predicted by this model for the speed to reach half its limiting value.

During the interval $0 \leq t \leq 6$, the pulling force from the engine has magnitude ct N, where c is a positive constant.

f) Show that $\frac{dv}{dt} + 0.12v = \frac{c}{m}t$.

g) Solve this differential equation to obtain v in terms of t for $0 \leq t \leq 6$.

h) Use your answers to show that $\frac{c}{m} = \frac{27e^{0.48} - 27}{125e^{0.72} - 35}$.

i) Find the total distance travelled by the carriage between $t = 0$ and $t = 10$. Give your answer to the nearest metre.

Question 44. A manufacturer tests the speed controller of an electric turntable. Let v be the angular speed (in suitable units) at time t hours. Initially the turntable is stationary, and in the long-term the speed approaches a constant value V.

a) A basic model assumes the rate of increase of speed is proportional to the difference between the limiting speed and the current speed. Write down a differential equation for the model.

b) Verify that $v = V(1 - e^{-kt})$, where $k > 0$, satisfies the differential equation, the initial condition, and the long-term condition.

c) A different model suggests that v satisfies the linear differential equation $\frac{dv}{dt} + \frac{2t}{1+t^2}v = \frac{2Vt}{1+t^2} + F(t)$, where $F(t)$ is a function of t. Find an integrating factor for this equation, simplifying it as far as possible.

d) Suppose $F(t) = 0$. Solve the differential equation and show that $v = \frac{Vt^2}{1+t^2}$.

Question 45. A brief spike in the output signal S (in suitable units) from a sensor is to be modelled. The spike begins at $t = 0$ seconds, and initially the output is zero. After 2 seconds, the output is measured as $S = 8$. When the spike is occurring, S is modelled by the differential equation $\frac{dS}{dt} = S - \frac{3t^2}{\sqrt{1+t^2}}$. The spike is assumed to last until the output is again zero.

a) Solve the differential equation to obtain an expression for S in terms of t.

b) Find the length of time that the spike lasts according to the model. Give your answer in exact form.

c) Find, in exact form, the rate of increase, $\frac{dS}{dt}$, at the end of the spike.

Question 46. A particle of mass 2 kg moves along a straight line with velocity v m s^{-1} at time t seconds. When $t = 0$, the velocity is -2 m s^{-1}. For $0 \leq t \leq 2$, the resultant force on the particle is $F = \frac{8}{t^2+2t+2} - \frac{2v}{(t+1)(t^2+2t+2)}$ N.

a) Form a differential equation for this model.

b) Find the first time t for which the particle is stationary.

2nd Order DEs: Homogeneous

Questions 1–34

Question 1. For each of the following differential equations:

- state its order,
- state whether it is linear,
- state whether it is homogeneous,
- determine whether an auxiliary equation can be formed.

a) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 4y = 0$
b) $y'' + y^2 = 0$
c) $3y'' - 5y' = 7$
d) $y''' + 2y' - y = 0$
e) $\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 2y = 0$
f) $\ddot{x} + 4\dot{x} + 4x = e^x$
g) $(y')^2 + y = 0$
h) $2\ddot{x} - 3\dot{x} + x = 0$

Question 2. It is given that $\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y = 0$. Verify that these are all particular solutions of the differential equation:

a) $y = e^{2x}$
b) $y = 4e^{2x}$
c) $y = e^{5x}$
d) $y = 7e^{5x}$
e) $y = e^{2x} + e^{5x}$
f) $y = 4e^{2x} + 7e^{5x}$

Question 3. It is given that $\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y = 0$. Let $y = e^{mx}$, where m is a constant.

a) Show that $m^2 - 7m + 10 = 0$.

The quadratic equation has roots m_1 and m_2.

a) Show that $y = Ae^{m_1 x} + Be^{m_2 x}$ is a solution to the differential equation, where A and B are constants.

Question 4. For each of the following differential equations, by forming and solving the auxiliary equation, find the general solution.

a) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$

b) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 35y = 0$

c) $\frac{d^2y}{dx^2} + 18\frac{dy}{dx} + 80y = 0$

d) $2\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$

e) $\frac{d^2y}{dx^2} - 4y = 0$

f) $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} = 0$

Question 5. Find the general solution to the following differential equations.

a) $\ddot{x} + 3\dot{x} - 28x = 0$

b) $4\ddot{x} + 4\dot{x} - 15x = 0$

Question 6. Solve the differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 0$ given that $y = 2$ when $x = 0$, and $y = 3$ when $x = \ln 2$.

Question 7. Solve the differential equation $2y'' - 7y' + 3y = 0$ given that $y = 5$ when $x = 0$, and $y = -238$ when $x = \ln 4$.

Question 8. It is given that $\dfrac{d^2y}{dx^2} - 6\dfrac{dy}{dx} + 9y = 0$. Verify that these are all particular solutions of the differential equation:

a) $y = e^{3x}$
b) $y = 2e^{3x}$
c) $y = xe^{3x}$
d) $y = 5xe^{3x}$
e) $y = e^{3x} + xe^{3x}$
f) $y = 2e^{3x} + 5xe^{3x}$

Question 9. It is given that $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$. Let $y = xe^{mx}$, where m is a constant.

a) Show that $m^2 - 6m + 9 = 0$.

The quadratic equation has a repeated root m.

a) Show that $y = Axe^{mx} + Be^{mx}$ is a solution to the differential equation, where A and B are constants.

Question 10. For each of the following differential equations, by forming and solving the auxiliary equation, find the general solution.

a) $\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0$

b) $\frac{d^2y}{dx^2} + 12\frac{dy}{dx} + 36y = 0$

c) $4\frac{d^2y}{dx^2} + 12\frac{dy}{dx} + 9y = 0$

d) $9\frac{d^2y}{dx^2} - 42\frac{dy}{dx} + 49y = 0$

Question 11. Find the general solution to the following differential equations.

a) $y'' + 26y' + 169y = 0$

b) $9y'' - 48y' + 64y = 0$

Question 12. Solve the differential equation $\ddot{x} - 10\dot{x} + 25x = 0$ given that $x = 0$ when $t = 0$, and $x = 4$ when $t = 1$.

Question 13. Solve the differential equation $9\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + y = 0$ given that $y = 3$ when $x = 3$, and $y = 4$ when $x = 6$.

Question 14. It is given that $\frac{d^2y}{dx^2} + 4y = 0$. You may assume that $\frac{d}{dx}(e^{ikx}) = ke^{ikx}$, where k is a constant. Verify that these are all particular solutions of the differential equation:

a) $y = e^{2ix}$
b) $y = e^{-2ix}$
c) $y = 2e^{2ix} - 3e^{-2ix}$

Question 15. Use $e^{i\theta} = \cos\theta + i\sin\theta$ to show that $Ae^{2ix} + Be^{-2ix}$ can be written in the form $C\cos 2x + D\sin 2x$, where C and D are real constants.

Question 16. It is given that $\dfrac{d^2y}{dx^2} + 4y = 0$. Verify that these are all particular solutions of the differential equation:

a) $y = \cos 2x$
b) $y = \sin 2x$
c) $y = 3\cos 2x$
d) $y = 5\sin 2x$
e) $y = \cos 2x + \sin 2x$
f) $y = 3\cos 2x + 5\sin 2x$

Question 17. It is given that $\frac{d^2y}{dx^2} + 9y = 0$. Let $y = e^{mx}$, where m is a constant.

a) Show that $m^2 + 9 = 0$.

The quadratic equation has complex conjugate roots.

a) Show that $y = A\cos 3x + B\sin 3x$ is a solution to the differential equation, where A and B are constants.

Question 18. For each of the following differential equations, by forming and solving the auxiliary equation, find the general solution.

a) $\frac{d^2y}{dx^2} + 25y = 0$

b) $\frac{d^2y}{dx^2} + 49y = 0$

c) $4\frac{d^2y}{dx^2} + 9y = 0$

Question 19. Find the general solution to the following differential equations.

a) $\ddot{x} + 8x = 0$

b) $9\ddot{x} + 25x = 0$

Question 20. Solve the differential equation $y'' + 100y = 0$ given that $y = 0$ when $x = 0$, and $y = 3$ when $x = \dfrac{\pi}{20}$.

Question 21. Solve the differential equation $16\dfrac{d^2y}{dx^2} + y = 0$ given that $y = 2$ when $x = \pi$, and $y = -3$ when $x = 2\pi$.

Question 22. It is given that $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 13y = 0$. Let $y = e^{mx}$, where m is a constant.

a) Show that $m^2 - 4m + 13 = 0$.

b) Hence show that the roots of this equation can be written in the form $a \pm bi$.

Question 23. It is given that $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$. Verify that these are all particular solutions of the differential equation:

a) $y = e^{(2+3i)x}$

b) $y = e^{(2-3i)x}$

c) $y = 5e^{(2+3i)x} + 2e^{(2-3i)x}$

Question 24. Use $e^{i\theta} = \cos\theta + i\sin\theta$ to show that $Ae^{(2+3i)x} + Be^{(2-3i)x}$ can be written in the form $e^{2x}(C\cos 3x + D\sin 3x)$, where C and D are real constants.

Question 25. It is given that $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$. Verify that these are all particular solutions of the differential equation:

a) $y = e^{2x}\cos 3x$
b) $y = e^{2x}\sin 3x$
c) $y = 4e^{2x}\cos 3x$
d) $y = 5e^{2x}\sin 3x$
e) $y = e^{2x}\cos 3x + e^{2x}\sin 3x$
f) $y = 4e^{2x}\cos 3x + 5e^{2x}\sin 3x$

Question 26. It is given that $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 29y = 0$. Let $y = e^{mx}$, where m is a constant.

a) Show that $m^2 - 4m + 29 = 0$.
b) Hence show that the roots of this equation can be written in the form $a \pm bi$.
c) Show that $y = e^{2x}(A\cos 5x + B\sin 5x)$ is a solution to the differential equation, where A and B are constants.

Question 27. For each of the following differential equations:

- Form the auxiliary equation,
- Solve the auxiliary equation,
- Write down the general solution.

a) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 5y = 0$

b) $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 13y = 0$

c) $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 25y = 0$

Question 28. Find the general solution to the following differential equations.

a) $y'' - 6y' + 73y = 0$

b) $16y'' - 80y' + 101y = 0$

Question 29. Solve the differential equation $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 10y = 0$ given that $y = 4$ when $x = 0$, and $y = 2$ when $x = \frac{\pi}{2}$.

Question 30. Solve the differential equation $\ddot{x} - 4\dot{x} + 13x = 0$ given that $x = 2$ when $t = 0$, and $\dot{x} = 1$ when $t = 0$.

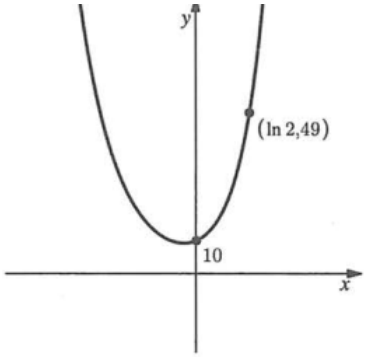
Question 31. Find the general solution to the following differential equations.

a) $\dfrac{d^2y}{dx^2} - 10\dfrac{dy}{dx} + 16y = 0$
b) $\dfrac{d^2y}{dx^2} - 18\dfrac{dy}{dx} + 97y = 0$
c) $\ddot{x} - 11\dot{x} - 60x = 0$
d) $121y'' - 110y' + 25y = 0$
e) $9\ddot{x} + 42\dot{x} + 49x = 0$
f) $\dfrac{d^2y}{dx^2} - 30\dfrac{dy}{dx} + 225y = 0$
g) $\dfrac{d^2y}{dx^2} + y = 0$
h) $2\ddot{x} + 2\dot{x} + 5x = 0$
i) $\dfrac{d^2y}{dx^2} + 6\dfrac{dy}{dx} + 109y = 0$
j) $81y'' + y = 0$
k) $\dfrac{d^2y}{dx^2} + 5\dfrac{dy}{dx} - 36y = 0$
l) $\dfrac{d^2y}{dx^2} + 16\dfrac{dy}{dx} + 64y = 0$
m) $49\ddot{x} + 25x = 0$
n) $30y'' + 49y' + 20y = 0$
o) $\dfrac{d^2y}{dx^2} + 400y = 0$
p) $\ddot{x} + 32x = 0$
q) $32y'' - 12y' - 35y = 0$

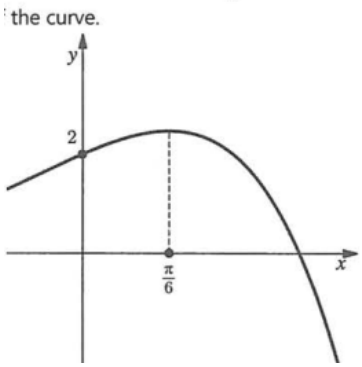
Question 32. For each of the following functions y, find the simplest 2nd order linear homogeneous differential equation with constant coefficients, for which the given function is a solution.

a) $y = 5e^{-3x} + 6e^{4x}$
b) $y = e^{6x}(2\cos 5x + 9\sin 5x)$
c) $y = 6xe^{14x} - 5e^{14x}$
d) $y = 15\cos^2 \frac{2}{3}x + 6\sin^2 \frac{2}{3}x$

Question 33. It is given that $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0$. A solution to the differential equation is shown in the diagram below. It is given that the curve passes through $(0, 10)$ and $(\ln 2, 49)$. Find the equation of the curve.



Question 34. It is given that $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$. A solution to the differential equation is shown in the diagram below. It is given that the curve passes through $(0, 2)$ and that the curve has a stationary point when $x = \frac{\pi}{6}$. Find the equation of the curve.



2nd Order DEs: Non-Homogeneous

Questions 1–43

Question 1. Given that $y = Ae^{-2x} + Be^{-3x}$, show that $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$.

a) Show that if $y = k$, where k is a constant, then $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 6k$.

b) Hence find a constant K such that $y = K$ is a solution of the differential equation $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 3$.

c) Show that $y = Ae^{-2x} + Be^{-3x} + K$ is a solution of the differential equation $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 3$.

Question 2. Find a particular integral for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 20y = 5$

b) $4\frac{d^2y}{dx^2} - 9y = 6$

c) $4\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 3y = -60$

Question 3. Using your answers to question 2, find the general solution for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 20y = 5$

b) $4\frac{d^2y}{dx^2} - 9y = 6$

c) $4\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 3y = -60$

Question 4. It is given that $\frac{d^2y}{dx^2} - \frac{dy}{dx} = 8$.

a) Show that a particular integral of the form $y = k$, where k is a constant, is unsuitable.

b) Show instead that a particular integral of the form $y = ax + b$, where a and b are constants, is suitable.

c) Hence find the general solution of the differential equation.

Particular Integrals of the form $y = ax + b$ where a and b are constants

Question 5. Find the values of a and b such that $y = ax+b$ is a solution of the differential equation $\frac{d^2y}{dx^2}+3\frac{dy}{dx}-10y = 10x + 7$.

Question 6. Find a particular integral for each of the following differential equations.

a) $\frac{d^2y}{dx^2} + 7\frac{dy}{dx} - 30y = 15x + 4$

b) $9\frac{d^2y}{dx^2} - y = 5 - x$

c) $4\frac{d^2y}{dx^2} + 16\frac{dy}{dx} - 9y = -45x$

Question 7. Using your answers to question 6, find the general solution for each of the following differential equations.

a) $\frac{d^2y}{dx^2} + 7\frac{dy}{dx} - 30y = 15x + 4$

b) $9\frac{d^2y}{dx^2} - y = 5 - x$

c) $4\frac{d^2y}{dx^2} + 16\frac{dy}{dx} - 9y = -45x$

Question 8. It is given that $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} = 8x$.

a) Show that a particular integral of the form $y = ax + b$, where a and b are constants, is unsuitable.

b) Show instead that a particular integral of the form $y = ax^2 + bx + c$, where a, b and c are constants, is suitable.

c) Hence find the general solution of the differential equation.

Particular Integrals of the form $y = ax^2 + bx + c$ where a, b and c are constants

Question 9. Find the values of a, b and c such that $y = ax^2 + bx + c$ is a solution of the differential equation $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 12x^2 + 6x - 3$.

Question 10. Find a particular integral for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} - 9y = 18x^2 + 11x - 14$

b) $4\frac{d^2y}{dx^2} - 16y = -8x^2$

c) $9\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 2y = 11 - 4x - 10x^2$

Question 11. Using your answers to question 10, find the general solution for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} - 9y = 18x^2 + 11x - 14$

b) $4\frac{d^2y}{dx^2} - 16y = -8x^2$

c) $9\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 2y = 11 - 4x - 10x^2$

Mix

Question 12. Find the general solution to each of the following differential equations.

a) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 4x + 1$

b) $\ddot{x} - 4\dot{x} + 4x = 4t^2 + 8t + 6$

c) $\frac{d^2y}{dx^2} + 9y = 9x$

d) $\ddot{x} - 5\dot{x} + 6x = 12$

e) $\frac{d^2y}{dx^2} + 4y = 8x^2$

f) $\ddot{x} + 4\dot{x} + 8x = 2t^2 + 4t$

g) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 4x + 2$

h) $25\ddot{x} + 10\dot{x} + x = 6 - t^2$

Question 13. Solve the differential equation $\frac{d^2y}{dx^2} + 25y = 50x^2 + 25x$ given that $y = 0$ when $x = 0$, and $\frac{dy}{dx} = 1$ when $x = 0$.

Question 14. Solve the differential equation $4\ddot{x} - 4\dot{x} + x = 4t + 5$ given that $x = 1$ when $t = 0$, and $\dot{x} = 0$ when $t = 0$.

Particular Integrals of the form $y = ke^{px}$ where k and p are constants

Question 15. Find the value of k such that $y = ke^x$ is a solution of the differential equation, where k is a constant: $\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 14y = 12e^x$.

Question 16. Find the value of k such that $y = ke^{2x}$ is a solution of the differential equation, where k is a constant: $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 12y = 8e^{2x}$.

Question 17. Find a particular integral for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 5e^{4x}$

b) $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 3e^{-2x}$

c) $2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 3y = 4e^{x}$

Question 18. Using your answers to question 17, find the general solution for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 5e^{4x}$

b) $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 3e^{-2x}$

c) $2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 3y = 4e^{x}$

Question 19.

a) Solve the differential equation $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 0$. It is given that $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 6e^{3x}$.

b) Substitute $y = ke^{3x}$ into the differential equation and explain why this trial function fails.

c) Find the value of k such that $y = kxe^{3x}$ is a solution of the differential equation.

d) Write down the general solution of the differential equation.

Question 20. Find the general solution to each of the following differential equations.

a) $2\ddot{x} - 5\dot{x} - 3x = 6e^{3t}$

b) $4\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + y = 5e^{x}$

c) $4\ddot{x} + 81x = 8e^{\frac{1}{2}t}$

d) $9\frac{d^2y}{dx^2} - 24\frac{dy}{dx} + 52y = 17e^{-x}$

Question 21. Solve the differential equation $\ddot{x} + 9x = 9e^{-2t}$ given that $x = 0$ when $t = 0$, and $\dot{x} = 9$ when $t = 0$.

Question 22. It is given that $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 12y = 60e^{-3x}$. Given also that $y \to 0$ as $x \to \infty$, and $\frac{dy}{dx} = 0$ when $x = 0$, solve the differential equation.

Question 23.

a) Solve the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$. It is given that $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 5e^{2x}$.

b) Substitute $y = ke^{2x}$ into the differential equation and explain why this trial function fails.

c) Substitute $y = kxe^{2x}$ into the differential equation and explain why this trial function fails.

d) Find the value of k such that $y = kx^2e^{2x}$ is a solution of the differential equation.

e) Write down the general solution of the differential equation.

Particular Integrals of the form $y = p\cos kx + q\sin kx$ where k, p and q are constants

Question 24. Find the values of the constants p and q such that $y = p\cos 3x + q\sin 3x$ is a solution of the differential equation $\frac{d^2y}{dx^2} + 10\frac{dy}{dx} + 9y = \sin 3x$.

Question 25. Find the values of the constants p and q such that $y = p\cos 4x + q\sin 4x$ is a solution of the differential equation $\dfrac{d^2y}{dx^2} + 9\dfrac{dy}{dx} - 36y = 5\cos 4x$.

Question 26. Find a particular integral for each of the following differential equations.

a) $25\dfrac{d^2y}{dx^2} - 9y = 9\cos 6x$

b) $2\dfrac{d^2y}{dx^2} - 11\dfrac{dy}{dx} + 5y = 89\sin 8x$

c) $3\dfrac{d^2y}{dx^2} + 11\dfrac{dy}{dx} + 6y = 20\sin 2x - 4\cos 2x$

Question 27. Using your answers to question 26, find the general solution for each of the following differential equations.

a) $25\dfrac{d^2y}{dx^2} - 9y = 9\cos 6x$

b) $2\dfrac{d^2y}{dx^2} - 11\dfrac{dy}{dx} + 5y = 89\sin 8x$

c) $3\dfrac{d^2y}{dx^2} + 11\dfrac{dy}{dx} + 6y = 20\sin 2x - 4\cos 2x$

Question 28.

a) Solve the differential equation $\dfrac{d^2y}{dx^2} + 4y = 0$. It is given that $\dfrac{d^2y}{dx^2} + 4y = \sin 2x$.

b) Substitute $y = p\cos 2x + q\sin 2x$ into the differential equation and explain why this trial function fails.

c) Find the values of p and q such that $y = px\cos 2x + qx\sin 2x$ is a solution of the differential equation.

d) Write down the general solution of the differential equation.

Mix

Question 29. It is given that $\frac{d^2y}{dx^2}+3\frac{dy}{dx}+2y=f(x)$, and the complementary function is $y=Ae^{-2x}+Be^{-x}$. Write down the trial function when:

a) $f(x)=5$
b) $f(x)=4e^{3x}$
c) $f(x)=2x+8$
d) $f(x)=3\sin x$
e) $f(x)=6x^2$
f) $f(x)=3e^{-x}$
g) $f(x)=8\cos 4x$
h) $f(x)=x$
i) $f(x)=5\cos 7x-9\sin 7x$
j) $f(x)=8e^{-2x}$

Question 30. It is given that $\frac{d^2y}{dx^2}+36y=f(x)$, and the complementary function is $y=A\cos 6x+B\sin 6x$. Write down the trial function when:

a) $f(x)=4e^{-x}$
b) $f(x)=1+2x$
c) $f(x)=3\cos 2x$
d) $f(x)=x^2-3$
e) $f(x)=6$
f) $f(x)=2\sin 6x$
g) $f(x)=7e^{6x}$
h) $f(x)=8\cos 6x$
i) $f(x)=2\sin 9x$
j) $f(x)=5\cos 6x-7\sin 6x$

Question 31. It is given that $\frac{d^2y}{dx^2} + 9\frac{dy}{dx} - 52y = f(x)$. Find the general solution for this differential equation when:

a) $f(x) = 12 - 13x$
b) $f(x) = 3e^{5x}$
c) $f(x) = 17e^{4x}$
d) $f(x) = 89\cos 3x + 89\sin 3x$

Question 32. It is given that $4\frac{d^2y}{dx^2} - 20\frac{dy}{dx} + 25y = f(x)$. Find the general solution for this differential equation when:

a) $f(x) = 841\cos x$
b) $f(x) = 18e^{x}$
c) $f(x) = 25 - 125x$
d) $f(x) = e^{\frac{5}{2}x}$

Prerequisite knowledge for Q33: Differentiation of Hyperbolic Functions

Question 33. It is given that $\ddot{x} + 2\dot{x} + 2x = f(t)$, where $x = 0$ when $t = 0$, and $\dot{x} = 0$ when $t = 0$. Find the particular solution to this differential equation when:

a) $f(t) = 2e^{-t}$
b) $f(t) = 5\sin t$
c) $f(t) = 6t^2$
d) $f(t) = 4\sinh t$
e) $f(t) = \cosh 2t - \sinh 2t$

Particular Integrals that are sums of polynomials, exponential functions or trigonometric functions

Question 34. Find the values of the constants p and q such that $y = pe^{2x} + qe^{4x}$ is a solution of the differential equation $5\dfrac{d^2y}{dx^2} + 9\dfrac{dy}{dx} - 2y = 4e^{2x} - 6e^{4x}$.

Question 35. Find the values of the constants p, q, r and k such that $y = p + qx + r\cos 2x + k\sin 2x$ is a solution of the differential equation $4\dfrac{d^2y}{dx^2} + y = 9x + \cos 2x$.

Question 36. Solve the differential equation $\dfrac{d^2y}{dx^2} - y = x + \sin x$.

Question 37. It is given that $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 42y = f(x)$, and the complementary function is $y = Ae^{-6x} + Be^{7x}$. Write down the trial function when:

a) $f(x) = 3 + e^x$
b) $f(x) = 3e^{2x} - e^x$
c) $f(x) = 6x - e^{5x}$
d) $f(x) = e^{4x} + 2\sin x$
e) $f(x) = \cos 3x - x^2$
f) $f(x) = 5 + 3x + e^{-5x}$
g) $f(x) = x - \sin \frac{1}{2}x - \cos \frac{1}{2}x$
h) $f(x) = e^{-6x} - 8$
i) $f(x) = e^{6x} + 8e^{7x} + x$
j) $f(x) = 5e^{-6x} - e^{7x} + \cos 3x$

Question 38. It is given that $\frac{d^2y}{dx^2} - 16\frac{dy}{dx} + 64y = f(x)$, and the complementary function is $y = Axe^{8x} + Be^{8x}$. Write down the trial function when:

a) $f(x) = \sin 3x - 7$
b) $f(x) = 3x - 10e^{-8x}$
c) $f(x) = e^x - 6\cos \frac{3}{2}x$
d) $f(x) = 5e^{4x} - 8 - 4x$
e) $f(x) = 9\sin 12x - 8 - 3\cos 12x$
f) $f(x) = 11e^{-3x} - e^{-2x} + 4x^2$
g) $f(x) = \sin x - \cos 2x$
h) $f(x) = 5x^2 + 3x - 8 + e^{10x}$
i) $f(x) = e^{8x} - 7x^2$
j) $f(x) = 2\cos 9x - 4\sin 4x - x^2 + 5e^{-8x}$

Question 39. Solve the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 3e^{2x} + 2e^{3x}$.

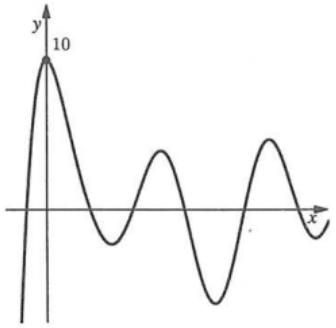
Question 40. Solve the differential equation $25\frac{d^2y}{dx^2} - 40\frac{dy}{dx} + 16y = 4 - 9e^{-\frac{2}{5}x} - 10e^{\frac{4}{5}x}$.

Question 41. It is given that $\ddot{x} - 7\dot{x} + 6x = f(t)$, where $x = 0$ when $t = 0$, and $\dot{x} = 1$ when $t = 0$. Find the particular solution for this differential equation when:

a) $f(t) = 72t + 74\sin t$
b) $f(t) = 50e^{t} - 52e^{2t}$
c) $f(t) = 35e^{6t} - 36t$
d) $f(t) = 108t^{2} - 100\cos 2t$

Question 42. Solve the differential equation $\dfrac{d^2y}{dx^2} + y = \sin x + \sin 2x$ given that $y = 5$ when $x = 0$, and $\dfrac{dy}{dx} = 3$ when $x = 0$.

Question 43. It is given that $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 17\cos 2x + 10\sin x$. A solution to the differential equation is shown in the diagram below. It is given that the curve has a maximum point at $(0, 10)$. Find the equation of the curve.



2nd Order DEs: Modelling

Questions 1–10

Question 1. The price $p(t)$, in pound sterling, of a particular type of fresh strawberries sold by a large chain of supermarkets is modelled over time t days after a sudden oversupply. Due to delays in adjusting pricing across stores and responding to customer demand, the price is assumed to satisfy

$$4\frac{d^2p}{dt^2} + 4\frac{dp}{dt} + p = 0.$$

At time $t = 0$, the price is £10 per kg and is decreasing at a rate of £2 per day.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Determine whether the price, according to the model, ever becomes negative.
d) Find the time at which the rate of decrease of the price is greatest.
e) Explain one limitation of this model in the context of supermarket pricing.

Question 2. A heavy fire door is released and begins to close automatically. Let $\theta(t)$ be the angular displacement, in radians, of the door from the fully closed position at time t seconds. The motion is modelled by

$$\frac{d^2\theta}{dt^2} + 5\frac{d\theta}{dt} + 6\theta = 0.$$

Initially, the door is held open at $\theta = 0.5$ and released from rest.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Find the time at which the angular speed of the door is greatest.
d) Find the angular velocity when $t = 1$.
e) Comment on the suitability of this model for large values of t.

Question 3. A carriage on a rotating unfair ride is displaced slightly from its equilibrium position. Let $\theta(t)$ be the angular displacement, in radians, from equilibrium at time t seconds. The motion is modelled by

$$9\frac{d^2\theta}{dt^2} + 6\frac{d\theta}{dt} + 2\theta = 0.$$

At time $t = 0$, $\theta = 0$ and the carriage is given an initial angular velocity of 3 rad s^{-1}.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Explain why the motion is oscillatory.
d) Find the maximum angular displacement after $t = 0$.
e) Find the second time at which the carriage comes instantaneously to rest.

Question 4. A water tank is part of a control system that regulates water level. Let $h(t)$, in metres, be the deviation of the water level from its equilibrium level at time t seconds. The system is modelled by

$$\frac{d^2h}{dt^2} + \frac{dh}{dt} + \frac{2}{9}h = 0.$$

At time $t = 0$, the water level is 0.3 m above equilibrium and is falling at a rate of 0.3 m s^{-1}.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Find the time at which the water level first reaches equilibrium.
d) Find the greatest deviation of the water level below equilibrium, and the time at which this occurs.
e) Find the rate of change of water level at $t = 3$.

Question 5. A camera on a stabilising tripod is displaced from its equilibrium position, where x m is the horizontal displacement from equilibrium at time t s. The motion is modelled by

$$\frac{d^2x}{dt^2} + a\frac{dx}{dt} + bx = 0,$$

where a and b are positive constants. At time $t = 0$, we have that $x = -0.02$, $\frac{dx}{dt} = 0$ and $\frac{d^2x}{dt^2} = 8$. It is given that the auxiliary equation has repeated roots.

a) Find the values of a and b.
b) Find the particular solution.
c) Describe the motion of the camera as time increases.

Question 6. A car is travelling along a straight road. Let $v(t)$, in m s^{-1}, be its speed at time t seconds. The motion is modelled by

$$\frac{d^2v}{dt^2} + \frac{3}{2}\frac{dv}{dt} + \frac{1}{2}v = 4.$$

Initially, the car has speed 0 m s^{-1} and acceleration 1 m s^{-2}.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Find the time at which the acceleration is greatest.
d) Find the limiting speed of the car.
e) Comment on the suitability of this model for large values of t.

Question 7. A room is being heated by a system whose output increases steadily over time. Let $T(t)$, in °C, be the temperature of the room at time t minutes. The temperature is modelled by

$$\frac{d^2T}{dt^2} + 3\frac{dT}{dt} + 2T = 17 + t.$$

At time $t = 0$, the temperature is 8°C and is increasing at 0.25°C per minute.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Show that the temperature is always increasing.
d) Find the temperature as $t \to \infty$.
e) Comment on the suitability of this model.

Question 8. A tank is being filled by a pump whose output increases over time, while water simultaneously drains from the tank. Let $h(t)$, in metres, be the height of water above a reference level at time t seconds. The system is modelled by

$$\frac{d^2h}{dt^2} + \frac{dh}{dt} + \frac{1}{4}h = t^2 + 2.$$

At time $t = 0$, it is given that $h = 0$ and $\frac{dh}{dt} = 0$.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Find $h''(0)$ and interpret what this means in context.
d) Determine whether, over time, the water level increases without bound.
e) Explain why this model may not be realistic for large values of t.

Question 9. A rotating platform at a fairground is subjected to a periodic driving force from its motor. Let $\theta(t)$, in radians, be the angular displacement from equilibrium at time t seconds. The motion is modelled by

$$80\frac{d^2\theta}{dt^2} + 40\frac{d\theta}{dt} + 25\theta = 136\cos\frac{1}{2}t.$$

At time $t = 0$, the platform is displaced by 0.4 radians from equilibrium and is rotating back towards equilibrium with angular velocity 0.5 rad s^{-1}.

a) Find θ in terms of t.
b) Describe the long-term behaviour of the motion.
c) Write the steady-state motion in the form $R\cos\left(\frac{1}{2}t - \alpha\right)$, where R and α are constants.
d) The platform is regarded as safe only if its steady-state angular displacement never exceeds 6.5 radians in magnitude. Determine whether the platform is safe.

Question 10. A chemical reactor is heated by an external process. The temperature $T(t)$, measured in °C above a reference level, changes over time as heat is supplied to the reactor. A model for the temperature is

$$36\frac{d^2T}{dt^2} + 108\frac{dT}{dt} + 81T = 216e^{-\frac{3}{2}t} + 100e^{\frac{1}{6}t},$$

where t is measured in minutes. Initially, the temperature is 4°C above the reference level and is decreasing at a rate of $\frac{1}{3}$ °C min^{-1}.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Describe the behaviour of the temperature as $t \to \infty$.
d) Estimate the temperature in the reactor after 10 minutes to 3 significant figures.
e) Estimate the time at which the temperature in the reactor is 20°C.

2nd Order DEs: Simple Harmonic Motion

Questions 1–39

Question 1. It is given that $x = 3\cos 2t - 4\sin 2t$.

a) Find $\dot{x}$.

b) Find $\ddot{x}$.

c) Show that $\ddot{x} = -\omega^2 x$, stating the value of ω.

Question 2. It is given that $x = A\cos 3t + B\sin 3t$.

a) Show that $\ddot{x} = -\omega^2 x$, stating the value of ω.

b) Given that $x = 2$ and $\dot{x} = -3$ when $t = 0$, find the values of A and B.

Question 3. Show that the general solution to the differential equation $\ddot{x} + \omega^2 x = 0$, where ω is a positive constant, is $x = A\cos\omega t + B\sin\omega t$.

Question 4. Solve the differential equation $\ddot{x} + \omega^2 x = 0$ given that $x = x_0$ and $\dot{x} = v_0$ when $t = 0$.

Question 5. Solve the following differential equations, subject to the given initial and boundary conditions.

a) Given that $\ddot{x} = -36x$, where $x = 3$ when $t = 0$ and $x = 4$ when $t = \frac{\pi}{12}$.

b) Given that $\ddot{x} = -400x$, where $x = -5$ when $t = 0$ and $x = 8$ when $t = \frac{3\pi}{40}$.

c) Given that $\ddot{x} = -2x$, where $x = 1$ when $t = 0$ and $\dot{x} = \frac{1}{3}$ when $t = 0$.

d) Given that $25\ddot{x} = -x$, where $x = \frac{1}{20}$ when $t = 0$ and $\dot{x} = \frac{1}{100}$ when $t = 10\pi$.

Question 6. Sketch the following graphs for $t \geq 0$.

a) $x = 3\cos t + 4\sin t$

b) $x = 4\cos 2t - 5\sin 2t$

c) $x = 4\sin 3t - 2\cos 3t$

Question 7. It is given that $\ddot{x} = -9x$, where $x = 2$ when $t = 0$ and $\dot{x} = 6$ when $t = \frac{\pi}{3}$.

a) Solve the differential equation.

b) Sketch a graph of x against t.

Question 8. It is given that $x(t) = A\cos\omega t + B\sin\omega t$.

a) Show that $x\left(t + \frac{2\pi}{\omega}\right) = x(t)$.

b) State the period of $x(t)$.

Question 9. It is given that $A\cos\omega t + B\sin\omega t \equiv R\cos(\omega t - \phi)$, where $R \geq 0$ and $0 < \phi < \frac{\pi}{2}$. Find the values of R and ϕ in terms of A and B.

Question 10. It is given that $\ddot{x} = -\omega^2 x$.

a) Explain what this says about the relationship between acceleration and displacement.

b) i) If $x > 0$, what is the sign of $\ddot{x}$?

ii) If $x < 0$, what is the sign of $\ddot{x}$?

iii) What does this imply about the motion relative to $x = 0$?

Question 11. A particle satisfies $\ddot{x} = -7x$.

a) State the relationship between acceleration and displacement.

b) Explain why $x = 0$ is an equilibrium position.

c) Describe the direction of motion immediately after passing through $x = 0$ with positive velocity.

Question 12. A particle satisfies $\ddot{x} = -\omega^2 x$. At some instant, $x > 0$ and $\dot{x} < 0$.

a) What is the sign of the acceleration?

b) Is the particle moving towards or away from equilibrium?

c) Explain whether the speed is increasing or decreasing.

Question 13. It is given that $x(t) = R\cos(\omega t - \phi)$.

a) Find $\dot{x}(t)$.

b) State the maximum speed.

c) Determine when turning points occur, in terms of $\omega t - \phi$, and the value of x at these points.

Question 14. A particle moves so that $x = R\cos(\omega t - \phi)$.

a) Show that when $t = \frac{\phi}{\omega}$, the particle is at maximum displacement.

b) Show that when $t = \frac{\phi}{\omega} + \frac{\pi}{2\omega}$, the particle passes through equilibrium.

c) Explain what the constant ϕ represents physically.

Question 15. A particle moves so that $x = R\cos\omega t$.

a) Find the velocity v of the particle in terms of t.

b) Using calculus, show that the maximum speed is ωR.

c) Explain why the maximum speed occurs at equilibrium.

Question 16. A particle moves along a line with displacement x and velocity $\dot{x}$ at time t.

a) Show that $\ddot{x} = \frac{dv}{dt}$.

b) Deduce that $\ddot{x} = v\frac{dv}{dx}$.

Question 17. A particle moves so that $\ddot{x} = -\omega^2 x$. By writing $\ddot{x}$ as $v\frac{dv}{dx}$, show that $v^2 + \omega^2 x^2 = c$ for some constant c.

Question 18. A particle moves so that $v^2 + \omega^2 x^2 = c$, where v is the velocity of the particle and x is the displacement over time. It is given that the amplitude of the motion is R.

a) Explain why $v = 0$ when $|x| = R$.

b) Use $x = \pm R$ and $v = 0$ to show that $v^2 + \omega^2 x^2 = \omega^2 R^2$.

Question 19. A particle moves so that $v^2 + 9x^2 = 36$.

a) Show that the motion is simple harmonic.
b) Find the angular frequency.
c) Find the amplitude.
d) Find the maximum speed.

Question 20. A particle moves so that $\ddot{x} = -25x$. Initially the particle is 0.2 m from equilibrium and is instantaneously at rest.

a) Find x as a function of t.
b) Find the period of x.
c) Find the maximum speed.
d) Find the first time the particle passes through equilibrium.

Question 21. A particle moves along a straight line and satisfies $\ddot{x} = -36x$. The greatest displacement from the centre is 0.5 m.

a) Find the period of the motion.

b) Find the greatest speed of the particle.

Question 22. A particle moves with SHM and satisfies $\ddot{x} = -4x$. Its maximum speed is 8 m s^{-1}.

a) Find the period of the motion.

b) Find the amplitude of the motion.

Question 23. A particle moves with SHM and satisfies $100\ddot{x} = -x$. When the particle is 0.3 m from the centre, its speed is 0.04 m s^{-1}.

a) Find the angular frequency of the motion.
b) Find the amplitude of the motion.
c) Find the period of the motion.
d) Find the speed when $x = 0.4$ m.

Question 24. A particle starts from the centre of oscillation with speed 5 m s^{-1}. It performs SHM with period $\frac{\pi}{2}$ seconds.

a) Find the angular frequency of the motion.
b) Find the amplitude of the motion.
c) Find the first time the particle reaches a displacement of 1 m.

Question 25. A particle moves so that $\ddot{x} = -16x + 32$, where displacement x is measured in metres, and time t is measured in seconds. At time $t = 0$, it is given that $x = 3$, and the particle is instantaneously at rest.

a) Find the equilibrium position.
b) Find the amplitude of the motion.
c) Find the period of the motion.

Question 26. A particle moves so that $\ddot{x} = -9x + 27$. At time $t = 0$, $x = 5$ and the particle is moving with speed 3 m s^{-1} towards the equilibrium position.

a) Find the equilibrium position.
b) Show that the motion is simple harmonic and find the angular frequency.
c) Find the amplitude of the motion.
d) Find the maximum speed of the particle.

Question 27. An offshore weather buoy is floating on the surface of the sea. It oscillates vertically in simple harmonic motion about its equilibrium level O. The motion has a period of 8π seconds and amplitude 1.5 metres. Initially the buoy is at O. Immediately after passing upward through O, the buoy rises to a point A which is 0.5 metres above O, and later reaches a point B which is 1.25 metres above O. Find the time taken for the buoy to travel from A to B.

Question 28. A camera mounted on a stabilising arm moves horizontally back and forth in simple harmonic motion about its central position C. When the camera is 0.4 metres from C, its speed is 1.2 m s^{-1}. When it is 0.8 metres from C, its speed is 0.6 m s^{-1}.

a) Find the amplitude of the motion.

b) Find the period of the motion.

Question 29. The end of a vibrating metal ruler moves with simple harmonic motion about a fixed point. When the tip is 15 mm from its central position, its speed is 120 mm s^{-1}. When it is 25 mm from the centre, its speed is 80 mm s^{-1}.

a) Find the amplitude of the vibration.

b) Find the period of the vibration.

Question 30. A cart of mass 2 kg moves along a smooth straight track. When it is x metres from a fixed point O, it experiences a force towards O of magnitude $18x$ N.

a) Form a differential equation for x in terms of t.

b) Show that the motion is SHM and find ω.

c) Find the period of the motion.

Question 31. A small trolley of mass 0.5 kg moves on a smooth horizontal surface. A control unit applies a force towards the point O of magnitude $2x$ N when the trolley is x metres from O. The trolley is pulled to $x = 0.3$ m and released from rest at $t = 0$ seconds.

a) Show that $\ddot{x} = -4x$.
b) Find x in terms of t.
c) Find the first time the trolley reaches O.
d) Find the speed of the trolley when $x = 0.2$ m.

Question 32. A light elastic spring of natural length 50 cm is fixed to a point O on a smooth horizontal table. A small block of mass 0.8 kg is attached to the other end. The block is pulled 0.25 m from its equilibrium position and released from rest. When the block is displaced x metres from equilibrium, the tension in the spring has magnitude $12x$ N and acts towards equilibrium.

a) Show that the motion is simple harmonic.
b) Find the period of the motion.
c) Find the maximum speed of the block.

Question 33. A block of mass 0.5 kg is attached to a light elastic spring on a smooth horizontal table. When the block is displaced x metres from its equilibrium position, the spring exerts a restoring force of magnitude $8x$ N. The block passes through equilibrium with speed 1.2 m s^{-1}.

a) Show that the motion is SHM and find ω.
b) Find the amplitude of the motion.
c) Find the maximum extension of the spring.

Question 34. A box of mass 2 kg slides on a smooth horizontal surface attached to a light elastic spring. The displacement x metres from equilibrium satisfies $\ddot{x} = -9x$. At time $t = 0$, the box is 0.5 metres from equilibrium and moving towards equilibrium with speed 1.5 m s^{-1}.

a) Find an expression for x in terms of t.
b) Find the amplitude of the motion.
c) Find the maximum speed of the box.

Question 35. A parcel of mass 1.2 kg rests on a smooth horizontal table and is attached to a light elastic spring fixed at one end. When the parcel is displaced x metres from equilibrium, the spring exerts a restoring force of magnitude $6x$ N. The parcel is pulled 0.6 m from equilibrium and released from rest.

a) Form a differential equation to model the motion of the parcel.
b) Solve the differential equation formed in (a) to find x in terms of t.
c) Find the time the parcel first returns to equilibrium.
d) State three modelling assumptions you have used to answer this question.

Question 36. A parcel of mass 1 kg rests at the midpoint O of two identical light elastic springs on a smooth horizontal surface. The other ends of the springs are attached to fixed points A and B, which lie on a straight line through O and are 3 metres apart. If the parcel is displaced x metres from O, each spring exerts a restoring force of magnitude $5x$ N towards O. The parcel is pulled 0.4 m towards B and released from rest.

a) Show that the motion is SHM.
b) Find the period of the motion.
c) Find the speed when the parcel is 0.2 metres from O.

Question 37. A small package of mass 2 kg lies on a smooth horizontal table at a midpoint O of two identical light elastic springs. The other ends of the springs are fixed at points A and B on a straight line, with $AO = BO = 1.5$ m. When the package is displaced x metres from O towards B, each spring exerts a force of magnitude $4x$ N on the package directed towards O. The package is pulled 0.6 m towards B and released from rest at $t = 0$.

a) Show that the package performs SHM and find ω.
b) Find the period of the motion.
c) Find the speed of the package when it is 1.1 metres from A.

Question 38. A small metal puck of mass m kg rests at a point O on a smooth horizontal air table. The puck is attached to two identical light springs along a straight line through O. The fixed ends of the springs are at points A and B, which are 2 m apart, with O the midpoint of AB. The puck is given a sharp tap so that it starts moving from O towards A with speed 0.8 m s^{-1}. At time t seconds, its displacement from O is x metres (take x to be positive in the direction from O towards A). Each spring exerts a force of magnitude $9mx$ N on the puck, directed towards O. The puck is modelled as a particle. The table is smooth and the only horizontal forces on the puck are due to the springs. Find the minimum distance from the puck to A during the subsequent motion.

Question 39. A small trolley of mass m kg moves on a smooth horizontal track. It is attached to two light elastic springs along a straight line. The left-hand spring is fixed to a wall at A, and the right-hand spring is fixed to a wall at B, where $AB = 2.4$ m. When the trolley is at a distance x metres from A, the left-hand spring exerts a force of magnitude $10mx$ N towards A, and the right-hand spring exerts a force of magnitude $5m(2.4 - x)$ N towards B. The trolley is released from rest at a position 0.3 m to the right of its equilibrium position.

a) Find the equilibrium position.

b) Show that the subsequent motion is simple harmonic and find ω.

c) Find, in terms of m, the greatest force exerted by the right-hand spring during the motion.

2nd Order DEs: Introducing Damping

Questions 1–14

Question 1. The following equations are each the solution of a different homogeneous second order linear differential equation with constant coefficients. In each case, determine whether the solution shows that the differential equation was modelling heavy damping (overdamping), critical damping, light damping (underdamping), or neither.

a) $x = 4e^{-t} + 2e^{-2t}$
b) $x = e^{-3t}(4\cos t - 3\sin t)$
c) $x = 7te^{-4t} + 4e^{-4t}$
d) $x = 2e^{-6t} + 5e^{3t}$
e) $x = e^{4t}(5\cos 2t + 8\sin 2t)$
f) $x = 9e^{-t} - 3te^{-t}$
g) $x = e^{-7t}\left(3\sin\frac{t}{2} - 4\cos\frac{t}{2}\right)$
h) $x = 7e^{-6t} - 10e^{-12t}$
i) $x = 8te^{t} - 3e^{t}$
j) $x = 5e^{-4t}\cos 6t$

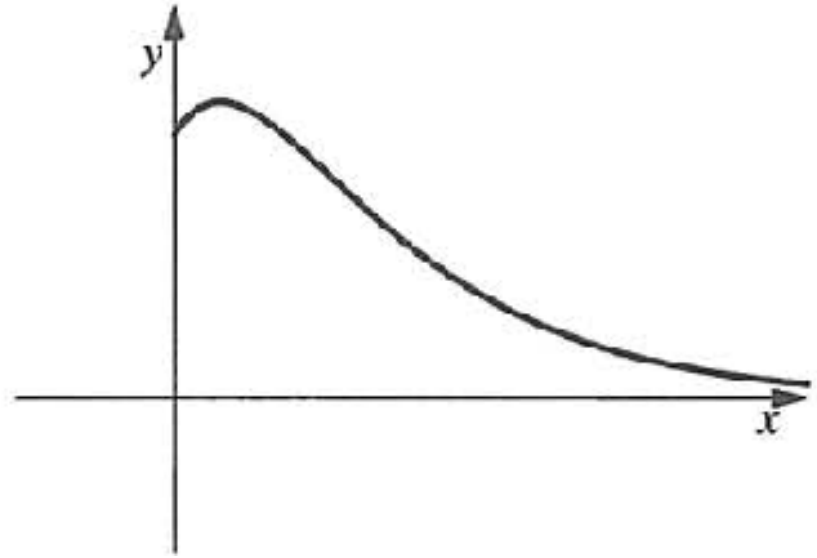
Question 2. Solve the following differential equations, classifying each scenario as heavy damping (overdamping), critical damping, or light damping (underdamping).

a) $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 5y = 0$
b) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$
c) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0$
d) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$
e) $\frac{d^2y}{dx^2} + 11\frac{dy}{dx} + 30y = 0$
f) $2\frac{d^2y}{dx^2} + 7\frac{dy}{dx} + 5y = 0$
g) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = 0$
h) $2\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 17y = 0$
i) $9\frac{d^2y}{dx^2} + 30\frac{dy}{dx} + 25y = 0$

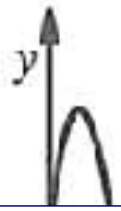
Question 3. The following graphs show solutions to differential equations of the form

$$\frac{d^2y}{dx^2} + k\frac{dy}{dx} + \omega^2 y = 0,$$

where $k^2 > 4\omega^2$ or $k^2 < 4\omega^2$. Describe what is happening in each scenario.



a)



Question 4. By finding the coordinates of any stationary points first, sketch the following graphs, given that $x \geq 0$.

a) $y = 3e^{-x} - 2e^{-2x}$
b) $y = 4e^{-3x} + 3e^{-x}$
c) $y = -3e^{-2x} - 6e^{-4x}$
d) $y = (2x - 1)e^{-2x}$

Question 5. Sketch the following graphs, given that $x \geq 0$.

a) $y = e^{-\frac{1}{2}x} \cos x$

b) $y = e^{-\frac{1}{3}x} \sin 3x$

c) $y = e^{-x}(\cos 2x + 2 \sin 2x)$

d) $y = e^{-3x}(-3 \cos 4x + \sin 4x)$

Question 6. It is given that $2\ddot{x} + 5\dot{x} + 3x = 0$, $t \geq 0$, where $x = 2$ and $\dot{x} = 1$ when $t = 0$.

a) Solve the differential equation to find x in terms of t.

b) Explain whether this is heavy damping (overdamping), critical damping, or light damping (underdamping).

c) Find the value of t when $\dot{x} = 0$.

d) Sketch the graph of x against t.

Question 7. It is given that $4\ddot{x} + 4\dot{x} + 5x = 0$, $t \geq 0$, where $x = 2$ and $\dot{x} = -1$ when $t = 0$.

a) Solve the differential equation to find x in terms of t.

b) Explain whether this is heavy damping (overdamping), critical damping, or light damping (underdamping).

c) State the period of the oscillations.

d) Sketch the graph of x against t.

Question 8. By finding the coordinates of any stationary points first, sketch the following graphs, given that $x \geq 0$.

a) $y = 4e^{-3x} - 2e^{-2x} + 1$

b) $y = e^{-2x} - 2e^{-3x} - 2$

c) $y = 3e^{-x} - 3e^{-2x} - 1$

d) $y = (3 - 9x)e^{-3x} + 2$

Question 9. Sketch the following graphs, given that $x \geq 0$.

a) $y = 3e^{-\frac{1}{2}x}\cos 2x + 3$

b) $y = e^{-\frac{1}{4}x}\sin x - 1$

c) $y = e^{-2x}(3\cos 4x - 2\sin 4x) - 3$

d) $y = e^{-x}(\cos 3x + 11\sin 3x) - 2$

Question 10. It is given that $\ddot{x} + 6\dot{x} + 8x = 8$, $t \geq 0$, where $x = 2$ and $\dot{x} = 2$ when $t = 0$.

a) Solve the differential equation to find x in terms of t.

b) Explain whether this is heavy damping (overdamping), critical damping, or light damping (underdamping).

c) Find the value of t when $\dot{x} = 0$.

d) Sketch the graph of x against t.

Question 11. It is given that $16\ddot{x} + 8\dot{x} + x = -3$, $t \geq 0$, where $x = 1$ and $\dot{x} = 1$ when $t = 0$.

a) Solve the differential equation to find x in terms of t.

b) Explain whether this is heavy damping (overdamping), critical damping, or light damping (underdamping).

c) Find the value of t when $\dot{x} = 0$.

d) Sketch the graph of x against t.

Question 12. It is given that $4\ddot{x} + 8\dot{x} + 13x = 26$, $t \geq 0$, where $x = 1$ and $\dot{x} = 4$ when $t = 0$.

a) Solve the differential equation to find x in terms of t.

b) Explain whether this is heavy damping (overdamping), critical damping, or light damping (underdamping).

c) State the period of the oscillations.

d) Sketch the graph of x against t.

Question 13. It is given that

$$\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 17y = 40\cos x, \qquad x \geq 0,$$

where $y = 2$ and $\frac{dy}{dx} = 2$ when $x = 0$.

a) Find a particular solution to the differential equation.

b) Describe what happens when x is large.

Question 14. It is given that

$$4\frac{d^2y}{dx^2} + 9y = \frac{9}{2}x, \qquad x \geq 0,$$

where $y = 4$ and $\frac{dy}{dx} = -1$ when $x = 0$.

a) Show that a particular solution to the differential equation is $y = 4\cos\frac{3}{2}x - \sin\frac{3}{2}x + \frac{1}{2}x$.

b) Find the smallest value of x such that y is stationary, giving your answer to three significant figures.

c) Sketch the graph of y against x.

The differential equation is modified so that the curve continues to oscillate about $y = \frac{1}{2}x$ but with reducing amplitude. The solution is now

$$y = e^{kx}\left(4\cos\frac{3}{2}x - \sin\frac{3}{2}x\right) + \frac{1}{2}x.$$

d) Explain why $k < 0$.

e) Show that $\left.\frac{dy}{dx}\right|_{x=0} \neq -1$.

f) Sketch the graph of y against x.

2nd Order DEs: Damped Motion

Questions 1–24

Question 1. A particle of mass 2 kg moves along a horizontal line. At time t, its displacement from a fixed point O is x metres, measured positive to the right. While P moves, two horizontal forces act on it: a restoring force of magnitude $10x$ N acting towards O and a resistance force of magnitude $6v$ N, where v m s^{-1} is the speed of P.

a) Write down the restoring force in terms of x, including its sign.

b) Write down the resistance force in terms of $\dfrac{dx}{dt}$, including its sign.

c) Show that the equation of motion is $\dfrac{d^2x}{dt^2} + 3\dfrac{dx}{dt} + 5x = 0$.

Question 2. A particle of mass 0.5 kg moves in a straight line. Its displacement from a fixed point A is x metres, measured positive away from A. At time t, the particle is moving with velocity v m s^{-1}. The forces acting on the particle include a force of magnitude $4x$ N acting towards A, and a resistance force of magnitude $2v$ N. Show that the equation of motion is

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 8x = 0.$$

Question 3. A particle P moves in a straight line. Its displacement from a fixed point O is x metres. The particle has mass m kg. Two forces act on P, including a force of magnitude kx N directed towards O, and a resistance force of magnitude λv N. Show that the equation of motion is

$$m\frac{d^2x}{dt^2} + \lambda\frac{dx}{dt} + kx = 0.$$

Question 4. A particle of mass 3 kg moves on the x-axis. Its displacement from the origin is x metres. At time t, three horizontal forces act on the particle, including: a restoring force of magnitude $12x$ N, a resistance force of magnitude $9v$ N, and a driving force of magnitude $15\cos t$ N acting in the positive x-direction. Show that the equation of motion is

$$\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 4x = 5\cos t.$$

Question 5. A particle P of mass 1 kg moves in a horizontal straight line. Its displacement from a fixed point O is x metres. A restoring force of magnitude $8x$ N acts on P towards O, and a resistance force of magnitude $4v$ N acts on P. Initially, the particle is at rest 2 metres from O.

a) Show that $\dfrac{d^2x}{dt^2} + 4\dfrac{dx}{dt} + 8x = 0$.

b) Find x as a function of t.

Question 6. A particle of mass 2 kg moves on the x-axis. Its displacement from the origin is x m. A restoring force of magnitude $6x$ N acts towards the origin, and a resistive force of magnitude $8v$ N acts on the particle. Initially, the particle is 3 metres from the origin and is travelling at 1 m s^{-1} towards the origin. Find x as a function of t.

Question 7. A particle P of mass 0.8 kg moves in a horizontal straight line. At time t seconds, its displacement from a fixed point O is x metres and its velocity is v m s^{-1}. A force of magnitude $5x$ N acts on P in the direction PO. The particle is also subject to a resistance force of magnitude $2v$ N. Initially, the particle is 1.2 metres from the origin and is travelling at 3 m s^{-1} in the direction of OP.

a) Show that the motion satisfies $4\dfrac{d^2x}{dt^2} + 10\dfrac{dx}{dt} + 25x = 0$.

b) Find the position of P when $t = 1$.

Question 8. A particle P of mass 1.5 kg moves along the x-axis with displacement from the origin is x m. Three forces act on P: a restoring force of magnitude $7.5x$ N towards the origin, a resistance force of magnitude $6v$ N, and a driving force $9\sin 2t$ N acting in the positive x-direction. Initially, $x = 4$ and the particle is travelling at 1 m s^{-1} away from the origin.

a) Show that the motion satisfies $\dfrac{d^2x}{dt^2} + 4\dfrac{dx}{dt} + 5x = 6\sin 2t$.

b) Find x as a function of t.

c) Describe the motion of P for large values of t.

Question 9. A mass attached to a vertical spring is released from rest after being displaced from its equilibrium position. The displacement x m from equilibrium at time t s satisfies

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 25x = 0.$$

Initially, the mass is pulled down 0.08 m and released from rest.

a) Find the displacement x as a function of t.
b) State whether the motion is oscillatory and justify your answer.
c) Determine the time of the first maximum displacement after release.

Question 10. The angular displacement θ radians of a hinged panel from its equilibrium position is modelled by

$$\frac{d^2\theta}{dt^2} + 2\frac{d\theta}{dt} + 10\theta = 0.$$

The panel is initially released from rest after being displaced 0.2 radians.

a) Find θ in terms of t.
b) State the type of damping in the system.
c) Describe the motion as $t \to \infty$.

Question 11. A particle of mass m kg is attached to the lower end of a light vertical spring. The particle moves only in the vertical direction. Its displacement from equilibrium at time t is x metres, measured downwards. While the particle is moving, two forces act on it in addition to its weight: a restoring force of magnitude $18x$ N acting towards the equilibrium position, and a resistive force of magnitude λv N, where v m s^{-1} is the speed of the particle. The particle is pulled down slightly and released.

a) Show that the motion is modelled by $m\dfrac{d^2x}{dt^2} + \lambda\dfrac{dx}{dt} + 18x = 0$.

b) Write down the auxiliary equation corresponding to the differential equation.

c) It is given that $m = 2$ and $\lambda = 8$.

i) Find the general solution for x.

ii) State whether the motion is oscillatory.

d) Keeping $m = 2$, determine the value of λ for which the motion is critically damped.

e) Using your value of λ from part (d), describe the motion for $m = 3$.

Question 12. A particle M of mass 2 kg is connected to a point C by a light elastic spring CD. The system lies on a smooth horizontal surface and is initially at rest with the spring at its natural length. At time $t = 0$, the point D is set in motion and moves away from C in a straight line with constant speed U. As the particle moves it experiences a resistive force proportional to its velocity. Let x denote the extension of the spring at time t. The motion of the system is modelled by the differential equation

$$2\frac{d^2x}{dt^2} + 7k\frac{dx}{dt} + 6k^2x = 4kU.$$

Find x in terms of t, k and U.

Question 13. A mass is attached to a vertical spring with a damper. The displacement of the mass from its equilibrium position is x cm at time t seconds after the motion begins. Upward displacement is taken as positive. The motion is modelled by the differential equation

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 4x = 10e^{-2t}.$$

In one experiment, the mass is initially positioned 18 cm above its equilibrium position and is given an initial velocity of 8 cm s^{-1} upwards.

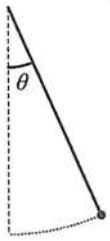
a) Find an expression for x in terms of t.
b) Find, to the nearest mm, the maximum displacement of the mass above its equilibrium position.
c) Determine whether the model predicts that the mass ever returns to its equilibrium position. Justify your answer.

Question 14. A particle of mass 1 kg is attached to a spring and moves along a horizontal line. Its displacement from equilibrium is x m at time t s. The particle experiences a resistive force proportional to its velocity and a small periodic driving force. The motion is modelled by

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 13x = 10\cos 2t.$$

a) Find an expression for x in terms of t.
b) Hence find the steady-state solution.
c) Find the amplitude of the steady-state solution.

Question 15. The angular displacement of a pendulum from the downward vertical is θ radians at time t seconds, as shown in the diagram below.



When air resistance and a small periodic driving force are taken into account, the motion of the pendulum is modelled by

$$\frac{d^2\theta}{dt^2} + 8\frac{d\theta}{dt} + 32\theta = 80\cos 4t.$$

a) Find the general solution of the differential equation.

Initially the pendulum is at the equilibrium position and is given an angular velocity of 9 rad s^{-1} in the positive direction.

b) Find an expression for θ in terms of t.

c) Describe the motion of the pendulum as t becomes large.

Question 16. A pendulum swings in a vertical plane about a fixed point. The angular displacement from the downward vertical is θ radians at time t seconds. Air resistance acts on the pendulum and a small magnetic impulse is applied when the motion begins. The motion is modelled by

$$\frac{d^2\theta}{dt^2} + 6\frac{d\theta}{dt} + 9\theta = 9e^{-t}.$$

Initially the pendulum is at the downward vertical and at rest. Show that, according to the model, the pendulum never returns to the downward vertical for any $t > 0$.

Question 17. A heavy door in a warehouse is fitted with a spring-loaded closing mechanism. When the door reaches the frame, it compresses a horizontal spring which slows the door and eventually brings it to rest. The door and its fittings have total mass M kg. When the spring is compressed by y metres, it exerts a force of magnitude $M\alpha^2 y$ N on the door. Let y metres be the compression of the spring t seconds after the door first makes contact with it.

a) Assuming no other horizontal forces act on the door, show that while the spring is being compressed the motion satisfies $\dfrac{d^2y}{dt^2} + \alpha^2 y = 0$.

b) The door first touches the spring with speed W. Write down expressions for y and $\dfrac{dy}{dt}$ when $t = 0$.

c) Find the compression of the spring as a function of time.

d) Find the maximum compression of the spring.

In reality, the closing mechanism contains oil which produces a resistive force proportional to the speed of the door. The magnitude of this force is $\lambda M v$ N, where v is the speed of the door and λ is a positive constant.

e) Show that the compression now satisfies $\dfrac{d^2y}{dt^2} + \lambda\dfrac{dy}{dt} + \alpha^2 y = 0$.

f) When $\lambda = 3\alpha$, determine whether the motion is Heavy Damped (overdamped), Critically Damped, or Lightly Damped (underdamped). Justify your answer.

g) When $\lambda = 3\alpha$, describe the motion of the door after it first touches the spring.

Question 18. A heavy door is fitted with a spring-loaded hinge and a hydraulic damping mechanism. When displaced from equilibrium, the hinge exerts a restoring moment proportional to θ, and the damping mechanism provides a resistive moment proportional to $\dfrac{d\theta}{dt}$. The angle θ radians through which the door is open at time t seconds is modelled by

$$\frac{d^2\theta}{dt^2} + 4\frac{d\theta}{dt} + 13\theta = 26.$$

Initially, the door is opened to an angle of 3 radians and released from rest.

a) Find the particular solution for θ.

b) Show that the solution may be written in the form $\theta = 2 + Re^{-2t}\cos(3t - \alpha)$ where R and α are constants.

c) State the limiting value of θ as $t \to \infty$, and explain its physical meaning.

d) Sketch the graph of x against t, showing clearly the equilibrium position.

Question 19. A crate of mass M kg is suspended from an elastic cable attached to a winch above it. The crate is allowed to drop vertically, and the cable first becomes taut at time $t = 0$. For t seconds after the cable first becomes taut, the extension of the cable is y metres, and the speed of the crate is v m s^{-1}. A model for the motion while the cable remains taut assumes that the forces acting on the crate are its weight, a tension in the cable of magnitude Cy N, and a resistive force of magnitude Dv N. Here C and D are positive constants, and it is given that $4CM > D^2$. Take downwards as the positive direction.

a) Show that the motion is modelled by the differential equation $M\dfrac{d^2y}{dt^2} + D\dfrac{dy}{dt} + Cy = Mg$.

b) Hence show that the general solution may be written in the form

$$y = e^{-\frac{D}{2M}t}\left(P\cos\left(\frac{\sqrt{4CM - D^2}}{2M}t\right) + Q\sin\left(\frac{\sqrt{4CM - D^2}}{2M}t\right)\right) + \frac{Mg}{C},$$

where P and Q are constants.

It is now given that $C = 25$, $D = 24$, $M = 9$, and $g = 10$, and that when the cable first becomes taut the extension is zero and the crate is moving downwards with speed 9 m s^{-1}.

c) Find the values of P and Q, and hence write down the corresponding expression for y in terms of t.

d) Show that the crate reaches its greatest extension when $\tan t = \frac{9}{2}$ and hence find the time at which the greatest extension occurs.

e) State the limiting value of y as $t \to \infty$, and explain what this means in the context of the motion.

A second model ignores the resistive force completely. The values of C, M and g remain unchanged.

f) Find the corresponding expression for y in terms of t.

g) Use the solutions from parts (c), (d) and (f) to show that the greatest extension predicted by the undamped model is larger than the greatest extension predicted by the damped model.

h) Sketch the expressions for y in terms of t for both the damped model and the undamped model on the same axes, identifying clearly which is which.

Question 20. Prerequisite knowledge for Q20: Mean Value of a Function

A particle Q moves along the x-axis. At time t seconds, its displacement from the origin O is x metres and its velocity is v m s^{-1}. The motion of Q is modelled by the differential equation $\ddot{x} + \lambda\dot{x} = 0$, where λ is a positive constant. Initially Q is projected from O with speed u in the positive x-direction.

a) Using $v = \dfrac{dx}{dt}$, show that the motion satisfies $\dfrac{dv}{dx} + \lambda = 0$.

b) Hence show that while the particle is moving, $v = u - \lambda x$.

The particle first comes to instantaneous rest at the point A, where $x = d$.

c) Show that $u = \lambda d$.

d) Hence prove that $v = \lambda(d - x)$.

e) The quantity z is defined by $z = \dfrac{1}{v}$. Show that the mean value of z with respect to the displacement as the particle moves from O to the point where $x = \frac{1}{2}d$ can be written as $\dfrac{\ln 4}{u}$.

Question 21. The motion of a particle is given by

$$x = 5e^{-0.2t}\cos 3t.$$

a) Find the amplitude of the motion at time t.
b) Find the ratio of the amplitudes at time t and $t + T$, where T is the period of the motion.

Question 22. A particle oscillates vertically about an equilibrium position. Its displacement from equilibrium is y cm at time t seconds. A model for the motion of the particle is

$$\frac{d^2y}{dt^2} + 2c\frac{dy}{dt} + 10y = 0,$$

where c is a positive constant.

a) Given that the motion is oscillatory, find the general solution.

It is observed experimentally that the amplitude of successive oscillations decreases so that the amplitude after each complete oscillation is 0.9 times the previous amplitude.

b) Show that $e^{-cT} = 0.9$, where T is the period of the oscillations.
c) Hence determine the value of c.

Initially the particle passes through equilibrium moving in the positive direction with speed 15 m s^{-1}.

d) Find an expression for $y(t)$, writing constants to three significant figures.

Question 23. A glider G of mass m kg moves along a straight horizontal air track through a fixed equilibrium point E. At time t seconds, the displacement of G from E is x metres and its velocity is v m s^{-1}. When G is displaced, the track's control mechanism exerts a restoring force of magnitude $4mx$ N directed towards E. Initially, the glider is 1 m to the right of E and is moving further to the right with speed 2 m s^{-1}.

a) Show that, while only the restoring force acts, the motion satisfies $\dfrac{d^2x}{dt^2} + 4x = 0$.

b) Find x in terms of t.

c) Find the amplitude and period of the motion.

d) Name the type of motion.

A resistive force of magnitude $2mv$ N is now introduced.

e) Show that the equation of motion becomes $\dfrac{d^2x}{dt^2} + 2\dfrac{dx}{dt} + 4x = 0$.

f) Determine the general solution of this differential equation.

g) State, with a reason, whether the motion is heavy damped (overdamped), critically damped, or light damped (underdamped).

Finally, while the resistive force is still present, a driving force of magnitude $4m\cos t$ N acts on the glider, so that the motion is modelled by

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 4x = 4\cos t.$$

At $t = 0$, the glider is again 1 m to the right of E and has velocity 2 m s^{-1} to the right.

h) Find $x(t)$.

i) Show that, for large values of t, the glider performs simple harmonic motion with period just over 6 seconds.

j) Find the amplitude of this long-term motion.

k) Show that the part of the solution in part (h) multiplied by e^{-t} can be written in the form $Ce^{-t}\cos(\sqrt{3}t - \beta)$ where $C > 0$.

l) Hence show that $x_s(t) - Ce^{-t} \le x(t) \le x_s(t) + Ce^{-t}$ where $x_s(t) = \frac{12}{13}\cos t + \frac{8}{13}\sin t$.

m) Sketch the graph of x against t, showing these bounding curves.

Question 24. A light horizontal spring is attached to a particle and a damping device. The particle is also acted on by a periodic driving force. Its displacement x m from equilibrium at time t s is found to be

$$x = e^{-\frac{1}{2}t}(A\cos 2t + B\sin 2t) + \cos t - \sin t,$$

where A and B are constants.

a) Show that x satisfies a differential equation of the form

$$p\frac{d^2x}{dt^2} + q\frac{dx}{dt} + rx = m\cos t + n\sin t,$$

where p, q, r, m and n are integers to be found.

b) Sketch a graph of x against t in the case where $A = 1$ and $B = 1$.

Coupled 1st Order Linear DEs

Questions 1–30

Question 1. It is given that $\frac{dy}{dt} = y$.

a) Solve this differential equation to find a general solution for y.

It is also given that $\frac{dx}{dt} = x + y$.

b) Using your result from part (a), solve this second differential equation to find a general solution for x.

Question 2. It is given that $\frac{dx}{dt} = 2x$.

a) Solve this differential equation to find a general solution for x.

It is also given that $\frac{dy}{dt} = x + 2y$.

b) Using your result from part (a), solve this second differential equation to find a general solution for y.

Question 3. It is given that $\frac{dx}{dt} = x + y$ and $\frac{dy}{dt} = 2y$.

a) Rearrange the first equation to express y in terms of x and $\frac{dx}{dt}$.

b) Hence, find an expression for $\frac{dy}{dt}$ in terms of x, $\frac{dx}{dt}$ and $\frac{d^2x}{dt^2}$.

c) By substituting your expressions from parts (a) and (b) into the second equation, show that $\frac{d^2x}{dt^2} - 3\frac{dx}{dt} + 2x = 0$.

d) Solve this differential equation to find a general solution for x.

e) Hence find a general solution for y in terms of t.

Question 4. It is given that $\frac{dx}{dt} = 2x + y$ and $\frac{dy}{dt} = x + 2y$.

a) Form a second-order differential equation in terms of x.

b) Solve this differential equation to find a general solution for x in terms of t.

c) Hence find a general solution for y in terms of t.

Question 5. It is given that $\frac{dx}{dt} = x + y$ and $\frac{dy}{dt} = -4x + 5y$.

a) Form a second-order differential equation in terms of y.
b) Solve this differential equation to find a general solution for y in terms of t.
c) Hence find a general solution for x in terms of t.

Question 6. It is given that $\frac{dx}{dt} = y$ and $\frac{dy}{dt} = -x + 2y$.

a) Form a second-order differential equation in terms of x.
b) Solve this differential equation to find the general solution for x in terms of t.
c) Hence find the general solution for y in terms of t.

Question 7. It is given that $\frac{dx}{dt} = x + 3y$ and $\frac{dy}{dt} = x - y$.

a) Form a second-order differential equation in terms of y.

b) Solve this differential equation to find the general solution for y in terms of t.

c) Hence find the general solution for x in terms of t.

It is also given that when $t = 0$, $x = 4$ and $y = 0$.

d) Find the particular solutions for x and y.

Question 8. Solve each of the following systems of differential equations to find general solutions for x and y in terms of t.

a) $\frac{dx}{dt} = 2x + y, \quad \frac{dy}{dt} = x + 2y$

b) $\frac{dx}{dt} = 2x + 2y, \quad \frac{dy}{dt} = -2x + 6y$

c) $\frac{dx}{dt} = x + 2y, \quad \frac{dy}{dt} = -x - y$

d) $\frac{dx}{dt} = 2x + y, \quad \frac{dy}{dt} = -5x$

Question 9. It is given that $\frac{dx}{dt} = x + y + 1$ and $\frac{dy}{dt} = x + y - 1$.

a) Form a second-order differential equation in terms of x.

b) Solve this differential equation to find the general solution for x in terms of t.

c) Hence find the general solution for y in terms of t.

It is also given that when $t = 0$, $x = 1$ and $y = 0$.

d) Find the particular solutions for x and y.

Question 10. Solve each of the following systems of differential equations to find general solutions for x and y in terms of t.

a) $\frac{dx}{dt} = 2y + 6, \quad \frac{dy}{dt} = 2x - 4$

b) $\frac{dx}{dt} = 3x + y + 2, \quad \frac{dy}{dt} = -x + y - 2$

c) $\frac{dx}{dt} = 2x + y + 3, \quad \frac{dy}{dt} = -13x - 2y - 5$

d) $\frac{dx}{dt} = 3x + 2y + 1, \quad \frac{dy}{dt} = -2x + 3y + 1$

Question 11. A system of differential equations is given by $\frac{dx}{dt} = x + y + t$ and $\frac{dy}{dt} = 2y + t^2$. It is also given that when $t = 0$, $x = \frac{7}{4}$ and $y = \frac{1}{4}$. Find expressions for x and y in terms of t.

Question 12. It is given that $\frac{dx}{dt} = 2x + y + \sin t$ and $\frac{dy}{dt} = -x + \cos t$.

a) Form a second-order differential equation in terms of x.

b) Solve this differential equation to find the general solution for x in terms of t.

c) Hence find the general solution for y in terms of t.

It is also given that when $t = 0$, $x = 1$ and $y = 1$.

d) Find the particular solutions for x and y.

Question 13. Solve each of the following systems of differential equations to find general solutions for x and y in terms of t.

a) $\dfrac{dx}{dt} = 2x + y + \sin t, \quad \dfrac{dy}{dt} = 2x + 3y + \cos t$

b) $\dfrac{dx}{dt} = x + y + \sin 4t, \quad \dfrac{dy}{dt} = -x + 3y + 3\sin 4t$

c) $\dfrac{dx}{dt} = 3x + 2y + \cos 2t, \quad \dfrac{dy}{dt} = -9x - 3y - 5\cos 2t$

d) $\dfrac{dx}{dt} = 2x + y + \sin 5t, \quad \dfrac{dy}{dt} = -x + 2y - 25\cos 5t + 22\sin 5t$

Question 14. A system of differential equations is given by $\dfrac{dx}{dt} = 2x + 3y + e^{-2t}$ and $\dfrac{dy}{dt} = x + 2y + e^{-2t}$. It is also given that $x(0) = \dfrac{25}{13}$ and $y(0) = -\dfrac{3}{13}$. Find expressions for x and y in terms of t.

Question 15. A system of differential equations is given by $\frac{dx}{dt} = -3x + 4y + e^{-t}$ and $\frac{dy}{dt} = \frac{3}{2}x - 4y + e^{-t}$. It is also given that $x(0) = 1$ and $y(0) = \frac{1}{4}$. Find expressions for x and y in terms of t. **Modelling: Predator Prey**

Question 16. On a nature reserve, rabbits and foxes interact. At time t years, x is the number of rabbits, and y is the number of foxes. The populations follow $\frac{dx}{dt} = 0.3x - 12y$ and $\frac{dy}{dt} = 0.02x - 1.5$.

a) Explain the meaning of the $0.3x$ and $-12y$ terms in the equation for $\frac{dx}{dt}$.

b) Explain the meaning of the $0.02x$ and -1.5 terms in the equation for $\frac{dy}{dt}$.

c) State what would happen to the rabbit population if there were no foxes present.

d) State what would happen to the fox population if there were no rabbits present.

e) Explain why the term -1.5, rather than $-1.5y$, might be unrealistic in a real biological model.

Question 17. In a lake, fish are preyed upon by birds. At time t years, x is the number of fish, and y is the number of birds. The system is modelled by $\frac{dx}{dt} = 0.5x - 8y$ and $\frac{dy}{dt} = 0.02x - 1.6$.

a) State the meaning of the constants 0.5, -8, 0.02, and -1.6.

b) Find the equilibrium point (where both populations are constant).

Question 18. On a tropical island, a population of insects has grown rapidly. To control them, frogs have been introduced. At time t years, where $t \geq 0$, x is the number of insects, and y is the number of frogs. Initially, there are 1200 insects and 25 frogs. For $t \geq 0$, it is assumed that:

- Insects reproduce at a rate of p% per year.
- Each frog eats, on average, q insects per year.
- Frogs die at a constant rate of r frogs per year.
- The frog population increases at a rate of s% of the insect population per year.

This leads to the system $\frac{dx}{dt} = 0.4x - 4y$ and $\frac{dy}{dt} = 0.02x - 2$.

a) State the values of p, q, r, and s.

b) Solve the coupled differential equations to find expressions for x and y in terms of t.

c) Given that $x > 0$ and $y > 0$ for $0 \leq t \leq 4$, use your solution to determine whether the insect population is being successfully controlled.

Question 19. In a woodland ecosystem, voles are preyed upon by owls. At time t years, x is the number of voles and y is the number of owls. It is known that the populations can be modelled by equations of the form $\frac{dx}{dt} = Ax + By$ and $\frac{dy}{dt} = Cx + Dy$. For $t > 0$, the following assumptions are made:

- In the absence of predators, the vole population decreases by 5% per year due to limited resources.
- Each owl causes the vole population to decrease by 2 voles per year.
- The owl population increases at a rate of 0.02 times the vole population per year.
- In the absence of food, the owl population decreases by 5% per year.

Initially, there are 300 voles and 20 owls.

a) State the values of A, B, C and D.

b) Explain what it means, in this model, for the owl death term to be proportional to y.

c) Solve the coupled differential equations to find expressions for x and y in terms of t.

d) Find the first time at which the vole population becomes zero. Explain why the model is no longer valid after this time.

Modelling: Two Tanks

Question 20. Tank 1 contains 500 litres of liquid. Salt solution with concentration 4 grams per litre flows into tank 1 at 8 litres per minute. The mixture in tank 1 is well stirred and flows out at 8 litres per minute. Let x be the amount of salt (in grams) in tank 1 at time t minutes.

a) Write an expression for the concentration of salt in the tank.

b) Find the rate at which salt leaves the tank.

c) Show that $\frac{dx}{dt} = 32 - \frac{2}{125}x$.

The liquid leaving tank 1 flows into tank 2, which also contains 500 litres of liquid. In addition, salt solution of concentration 10 grams per litre flows directly into tank 2 at 2 litres per minute. The mixture in tank 2 is well stirred and flows out of tank 2 at 10 litres per minute. Let y be the amount of salt in tank 2 at time t minutes.

d) Write down a differential equation for $\frac{dy}{dt}$ in terms of x and y.

Question 21. Two tanks, A and B, each contain 400 litres of liquid. Liquid flows from A to B at 12 litres per minute. Liquid leaves B at 12 litres per minute. Initially, tank A contains 60 grams of salt and tank B contains none. The liquid in each tank is well-stirred at all times. Let x be the amount of salt (in grams) in tank A at time t minutes, and let y be the amount of salt (in grams) in tank B at time t minutes.

a) Write down an expression for $\dfrac{dx}{dt}$.

b) Show that $\dfrac{dy}{dt} = \dfrac{3}{100}x - \dfrac{3}{100}y$.

c) Solve the coupled differential equations to find expressions for x and y in terms of t.

Question 22. Two tanks, P and Q, each contain 700 litres of liquid. Liquid flows from P to Q at 15 litres per minute, and liquid flows from Q to P at 7 litres per minute. No liquid flows into P or Q from outside. Liquid leaves Q at a rate that keeps volume constant. Let x and y be the amounts of salt in P and Q.

a) Explain why liquid must leave Q at 8 litres per minute.

b) Write down expressions for $\dfrac{dx}{dt}$ and $\dfrac{dy}{dt}$.

Question 23. Two tanks, M and N, each contain 900 litres of liquid. Salt solution of concentration c grams per litre flows into M at 7 litres per minute. Liquid flows from M to N at 16 litres per minute, and liquid flows from N to M at 9 litres per minute. Liquid leaves N so that volumes remain constant. Let x and y be the grams of salt in M and N.

a) Show that $\frac{dx}{dt} = 7c - \frac{4}{225}x + \frac{1}{100}y$.

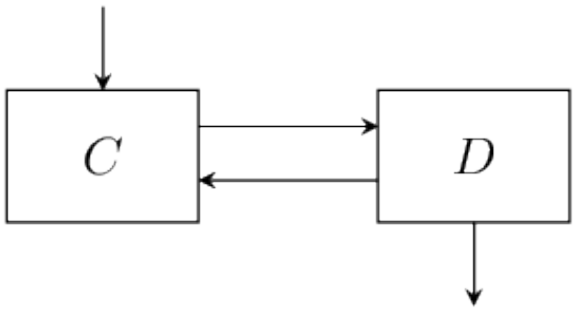
b) Write down a differential equation for y.

c) Given that the system reaches a steady state where $x = 450$ and $y = 450$, find the value of c.

It is given that initially there are 500 grams of salt in tank M and 400 grams of salt in tank N.

d) Solve the coupled differential equations to find expressions for x and y in terms of t.

Question 24. Two large containers, C and D, are each filled with 1000 litres of liquid. At time $t = 0$, both containers contain pure water.



For $t > 0$, the system operates as follows:

- A solution containing p grams per litre of salt flows into container C at a constant rate.
- Liquid flows from container C to container D at 25 litres per minute.
- Liquid flows from container D to container C at 9 litres per minute.
- Liquid leaves container D through an outlet.
- The volume of liquid in each container remains constant.

Let $x(t)$ be the mass of salt, in grams, in container C, and let $y(t)$ be the mass of salt, in grams, in container D.

a) Show that the rate at which liquid flows into container C is 16 litres per minute.

b) Show that the system can be modelled by $\frac{dx}{dt} = 16p - 0.025x + 0.009y$ and $\frac{dy}{dt} = 0.025x - 0.025y$.

c) Given that $p = 4$, solve the system of differential equations to find expressions for $x(t)$ and $y(t)$.

d) Find the limiting values of x and y as $t \to \infty$, and interpret your answer in terms of the concentrations in the two containers.

Modelling: Mix

Question 25. Two substances, A and B, are involved in a reaction inside a closed vessel. Let $a(t)$ and $b(t)$ denote the masses (in grams) of substances A and B respectively, at time t minutes after the reaction begins. The system is modelled by the differential equations $\frac{da}{dt} = -3a + 4b + 6$ and $\frac{db}{dt} = -2a + 2b + 10$.

a) Show that a satisfies $\frac{d^2a}{dt^2} + \frac{da}{dt} + 2a = 28$.

b) Solve this differential equation to obtain the general solution for $a(t)$.

c) Hence find an expression for $b(t)$.

Initially, A has a mass of 4 grams and B has a mass of 1 gram.

d) Find the particular solutions for both $a(t)$ and $b(t)$.

e) Show that the system approaches a steady state as $t \to \infty$, and find the limiting values of $a(t)$ and $b(t)$.

f) A chemist claims that the reaction has effectively finished after 6 minutes. Using your model, comment on this claim, and state whether the system continues to move closer to equilibrium after this time.

g) Explain how the constant terms in the differential equations influence the long-term behaviour of the system.

Question 26. An ecologist is studying the interaction between two mammal species in a woodland: pine martens and red squirrels. Let $p(t)$ be the number of pine martens and $r(t)$ be the number of red squirrels at time t years. Their populations are modelled by the system $\frac{dp}{dt} = 2p - 2r - 341e^{-t}$ and $\frac{dr}{dt} = 4p - 4r$.

a) Show that p satisfies $\frac{d^2p}{dt^2} + 2\frac{dp}{dt} = -1023e^{-t}$.

b) Solve this differential equation to obtain the general solution for $p(t)$.

c) Hence find an expression for $r(t)$.

d) Initially, there are 2046 pine martens and 3411 red squirrels. Find the particular solutions for $p(t)$ and $r(t)$.

e) Determine the value of $t = T$ for which the pine marten population becomes zero. Give your answer to 3 decimal places.

f) Describe the long-term behaviour of the red squirrel population as $t \to \infty$.

Question 27. A pair of interacting quantities $x(t)$ and $y(t)$ are monitored in a system over time t. The following model is proposed:

- The rate of change of x is proportional to the difference between x and y.
- The rate of change of y is proportional to its current value but is reduced by an amount proportional to x.

This leads to the system $\dfrac{dx}{dt} = a(x - y)$ and $\dfrac{dy}{dt} = ay - bx$, where a and b are positive constants.

a) Show that x satisfies a second-order differential equation of the form $\dfrac{d^2x}{dt^2} - 2a\dfrac{dx}{dt} + a(a - b)x = 0$.

b) Hence show that the general solution for x can be written as $x(t) = Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t}$.

c) Using your result from part (b), find the general solution for $y(t)$ in terms of A, B, a, b and t.

d) Show that there exist constants K_1 and K_2 such that $x(t) + \sqrt{\dfrac{a}{b}}y(t) = K_1e^{(a-\sqrt{ab})t}$ and $x(t) - \sqrt{\dfrac{a}{b}}y(t) = K_2e^{(a+\sqrt{ab})t}$.

e) Assuming that $K_2 \neq 0$, determine $\lim\limits_{t\to\infty} \dfrac{y(t)}{x(t)}$.

f) A scientist claims: "Regardless of the initial values of x and y, the system eventually settles into a fixed ratio between the two quantities." Discuss whether this claim is valid under the model, making your reasoning clear.

Modelling: 3 Variables

Question 28. A chemical process involves three substances: A, B, and C. Substance A is converted into B, and B is converted into C. No material is lost during each stage of the process. The amounts (in kg) at time t hours are a, b, and c, respectively. It is known that:

- A decreases at a rate proportional to its amount.
- B increases due to conversion from A, and decreases due to conversion into C.
- C is formed from B.

a) Write down differential equations for $\dfrac{da}{dt}$ and $\dfrac{db}{dt}$, introducing suitable constants.

b) Explain why a third differential equation for $\dfrac{dc}{dt}$ is not independent.

Question 29. In a production process, material P is converted into Q, and then into R. No material is lost during each stage of the process. The amounts (in kg) at time t hours are p, q, and r. The system is modelled by $\frac{dp}{dt} = -2p$ and $\frac{dq}{dt} = 2p - q$. Initially, $p = 10$, $q = 0$, and $r = 0$.

a) Show that $p = 10e^{-2t}$.

b) Show that $\frac{dp}{dt} + \frac{dq}{dt} + \frac{dr}{dt} = 0$.

c) Explain what the result in part (b) means in the context of the process.

d) Find an expression for q in terms of t.

e) Find an expression for r in terms of t.

Question 30. A substance X is processed in stages: it is converted into Y, and then into Z. No material is lost during each stage of the process. The amounts (in kg) at time t hours are x, y, and z. The system is modelled by $\frac{dx}{dt} = -\frac{3}{100}x$ and $\frac{dy}{dt} = \frac{3}{100}x - \frac{1}{100}y$. Initially, $x = 100$, $y = 0$, and $z = 0$.

a) Find x in terms of t.

b) By differentiating the equation for $\frac{dy}{dt}$ and eliminating x, show that y satisfies the second-order differential equation $\frac{d^2y}{dt^2} + \frac{1}{25}\frac{dy}{dt} + \frac{3}{10000}y = 0$.

c) Find an expression for y in terms of t.

d) Find an expression for z in terms of t, and hence show that the amount of Z present after a long time is 100 kg.